
p -ADIC HAHN SERIES WITH SPARSE SUPPORT

by

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Abstract. — Let p be a prime number. We introduce a sparseness condition on the supports of p -adic Hahn series, and prove that this condition implies transcendence over $\check{\mathbf{Q}}_p$, the completed maximal unramified extension of $\mathbf{Q}_p$. As an application, we prove the order type conjecture of $\mathbf{Q}_p$ -algebraic p -adic Hahn series with bounded support under the condition that the support has only finitely many accumulation points.

All results in this paper have been fully formalized in Lean 4 with Mathlib.

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1. Introduction

Let p be a prime number. Let $\mathbf{F}_p$ be the finite field of p elements, $\overline{\mathbf{F}}_p$ be an algebraic closure of $\mathbf{F}_p$. Let $\mathbf{Q}_p$ be the field of p -adic numbers, $\check{\mathbf{Q}}_p = W(\overline{\mathbf{F}}_p)[p^{-1}]$ be the completed maximal unramified extension of $\mathbf{Q}_p$, $\check{\mathbf{Z}}_p = W(\overline{\mathbf{F}}_p)$ be the ring of integers of $\check{\mathbf{Q}}_p$, $\overline{\mathbf{Q}}_p$ be an algebraic closure of $\mathbf{Q}_p$, and $\mathbf{C}_p = \widehat{\overline{\mathbf{Q}}_p}$ be the field of p -adic complex numbers. We normalize the p -adic valuation on $\mathbf{C}_p$ and its subfields by $v_p(p) = 1$. For any set X , we denote by $\text{card}(X)$ the cardinality of X .

1.1. p -adic transcendence via p -adic Hahn series. — A p -adic Hahn series is a generalized formal power series of the form $f = \sum_{q \in \mathbf{Q}} [f(q)]p^q$, where $f(q) \in \overline{\mathbf{F}}_p$ and $[\cdot]$ is the Teichmüller lift, whose support $\text{Supp}(f) = \{q \in \mathbf{Q} \mid f(q) \neq 0\}$ is a well-ordered subset of $\mathbf{Q}$. Krull, Lampert, and Poonen showed that the set of p -adic Hahn series, which we denote by $\mathbf{L}_p$, forms the spherical completion of $\overline{\mathbf{Q}}_p$. In particular, $\mathbf{L}_p$ is algebraically closed and complete with respect to the p -adic valuation given by $f \mapsto \min \text{Supp}(f)$.

The field $\mathbf{L}_p$ provides a natural setting in which to study transcendental number theory over p -adic fields. More precisely, a fundamental question arises:

Question 1.1. — *Given a p -adic Hahn series $f \in \mathbf{L}_p$, how can one determine whether f is a p -adic algebraic number, i.e., algebraic over $\mathbf{Q}_p$?*

Although this question remains open in general, several necessary conditions for a p -adic Hahn series to be algebraic over $\mathbf{Q}_p$ are known:

1. In [Lam86] and [Poo93], Lampert and Poonen proved that if $f \in \mathbf{L}_p$ is algebraic over $\mathbf{Q}_p$, then
 - (a) there exists an integer T such that $\text{Supp}(f) \subset \frac{1}{T}\mathbf{Z}[1/p]$;
 - (b) there exists a finite extension $\mathbf{F}_q$ of $\mathbf{F}_p$ such that $\{f(q)\}_{q \in \mathbf{Q}} \subset \mathbf{F}_q$.

These conditions are also studied quantitatively in [WY24].

2. In [Lam86], Lampert also proved that if $f \in \mathbf{L}_p$ is algebraic over $\mathbf{Q}_p$, then the accumulation points of $\text{Supp}(f)$ are rational numbers.
3. In [Ked01b] and [Ked17], Kedlaya gives a necessary and sufficient condition for an equal-characteristic Hahn series in $\mathbf{L}_p^\flat := \overline{\mathbf{F}}_p((t^\mathbf{Q}))$ to lie in the algebraic closure $\overline{\mathbf{F}}_p((t))^\text{alg}$ of $\overline{\mathbf{F}}_p((t))$, phrased in the language of automata theory (cf. Theorem 6.1). As an application, Kedlaya uses Witt vectors to lift this result to the p -adic case (cf. Theorem 6.4): he shows that the field $\mathbf{C}_p$, when viewed as a subfield of $\mathbf{L}_p$, coincides with the completion of the set $\Theta(\overline{\mathbf{F}}_p((t))^\text{alg}, \wedge)$, where $\overline{\mathbf{F}}_p((t))^\text{alg}, \wedge$ is the t -adic completion of $\overline{\mathbf{F}}_p((t))^\text{alg}$ and $\Theta: \overline{\mathbf{F}}_p((t^\mathbf{Q})) \longrightarrow \mathbf{L}_p$ is the map $\sum_{q \in \mathbf{Q}} f(q)t^q \longmapsto \sum_{q \in \mathbf{Q}} [f(q)]p^q$.

Intuitively, Kedlaya's result indicates that the algebraicity of equal-characteristic Hahn series in $\overline{\mathbf{F}}_p((t^\mathbf{Q}))$ and the algebraicity of p -adic Hahn series in $\mathbf{L}_p$ are related to some extent via the map Θ , which leads to the following questions:

qu:29274

Question 1.2. —

1. Suppose that $f \in \mathbf{L}_p^\flat$ is algebraic over $\mathbf{F}_p((t))$. Is $\Theta(f)$ algebraic over $\mathbf{Q}_p$?
2. Suppose that $f \in \mathbf{L}_p$ is algebraic over $\mathbf{Q}_p$. Is $\Theta^{-1}(f)$ algebraic over $\mathbf{F}_p((t))$?

In [WY25], we gave a negative answer to the first question by showing that the p -adic Hahn series $\mathfrak{A} := \sum_{k=1}^{\infty} p^{-1/p^k}$ is transcendental over $\mathbf{Q}_p$, whereas its preimage under Θ is a root of the polynomial $X^p - X - t^{-1}$ over $\mathbf{F}_p((t))$.

1.2. Sparseness and main theorem. — The key observation in [WY25] is that the support of $\mathfrak{A}$ is “sparse”, in the sense that for any nonzero polynomial $P(X) \in \mathbf{Q}_p[X]$, the multinomial expansion of $P(\mathfrak{A})$ contains orphaned terms that cannot be cancelled by other terms, which forces $P(\mathfrak{A})$ to be nonzero. In this article, we generalize this idea to the following combinatorial definitions:

def:25951

Definition 1.3. —

1. Any rational number q can be uniquely written in the form $q = w + \sum_{i=1}^{\infty} q_i \cdot p^{-i}$, where $w \in \mathbf{Z}$, $q_i \in \{0, 1, \dots, p-1\}$ and $q_i \neq p-1$ for infinitely many i . We call $\mathfrak{N}_p(q) := \sum_{i=1}^{\infty} q_i \in \mathbf{N} \cup \{\infty\}$ the *p -digit sum* of q .
2. For any set S of rational numbers, define the *dominant p -digit sum* of S to be

$$\text{dom}_p(S) := \sup\{\mathfrak{N}_p(q) \mid q \in S\} \in \mathbf{N} \cup \{\infty\},$$

and define the *p -digit dominant part* of S to be the following subset of S :

$$\text{Dom}_p(S) := \{q \in S \mid \mathfrak{N}_p(q) = \text{dom}_p(S)\}.$$

def:15658

Definition 1.4. — A set $S \subset [0, 1) \cap \mathbf{Q}$ is *sparse* if:

1. $\text{dom}_p(S)$ is finite;
2. for infinitely many integers $n \geq 1$, there exist n elements $d_1, \dots, d_n \in \text{Dom}_p(S)$ such that

- (a) there is no carry in base p when adding $d_1, \dots, d_n$ together;
- (b) if $e_1, \dots, e_n \in S$ satisfy that $d_1 + \dots + d_n$ and $e_1 + \dots + e_n$ differ by an integer, then up to a permutation of $\{1, 2, \dots, n\}$, $d_i = e_i$ for every $i = 1, 2, \dots, n$.

Remark 1.5. — We give some comments on Definition 1.4:

1. The first condition is closely related to Kedlaya's criterion for the algebraicity of equal-characteristic Hahn series: he shows that if $f \in \mathbf{L}_p^b$ is algebraic over $\overline{\mathbf{F}}_p((t))$, then $\text{dom}_p(-T \cdot \text{Supp}(f))$ is finite for some integer $T \geq 1$.
2. Condition (2a) indicates that the addition $d_1 + \dots + d_n$, no matter how one parenthesizes it, behaves like the addition in the free commutative monoid $\bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$.
3. Condition (2b) is a combinatorial rigidity condition asserting that the chosen carry-free sum admits a unique decomposition modulo $\mathbf{Z}$. This produces the orphaned exponents in the multinomial expansion, which is the key mechanism behind the transcendence proof.

The following example illustrates a typical situation in which the sparseness condition is satisfied:

eg:61947

Example 1.6 (cf. Example 3.7). — Let $A_1, A_2, \dots$ be a family of pairwise disjoint nonempty subsets of $\mathbf{Z}_{\geq 1}$ such that $\sup_i \text{card}(A_i) < \infty$, and this supremum is attained by infinitely many i . Then the set

$$\left\{ \sum_{r \in A_i} p^{-r} \mid i = 1, 2, \dots \right\}$$

is sparse.

The main theorem of this article is the following:

thm:main

Theorem 1.7 (cf. Theorem 5.3). — Let $f \in \mathbf{L}_p$ be a p -adic Hahn series such that there exists an integer $T \geq 1$ for which $-T \cdot \text{Supp}(f)$ admits a sparse set $W \neq \{0\}$ of representatives modulo $\mathbf{Z}$. Then f is transcendental over $\mathbf{Q}_p$, and hence over $\mathbf{Q}$.

Remark 1.8. — The sparseness condition does not involve the coefficients of f , so it is natural that it does not distinguish algebraicity over $\mathbf{Q}_p$ from algebraicity over $\mathbf{Q}$.

1.3. Application: Order type conjecture of $\mathbf{Q}_p$ -algebraic p -adic Hahn series. — In [Ked01a, Section 4], Kedlaya proved that the order type of the support of a p -adic Hahn series that is algebraic over $\mathbf{Q}_p$ is at most ω^ω , where ω is the first infinite ordinal. However, the lack of criteria for the $\mathbf{Q}_p$ -algebraicity of p -adic Hahn series makes it difficult to determine which order types $\leq \omega^\omega$ can be realized as the order type of the support of a $\mathbf{Q}_p$ -algebraic p -adic Hahn series. In the same article, Kedlaya predicts that only finite order types, ω , and ω^ω are realizable. The following conjecture was formulated in our previous work, where it was shown to be implied by Kedlaya's prediction:

conj:47376

Conjecture 1.9 (cf. [WY25, Conjecture 5.1]). — Let $f \in \mathbf{L}_p$ be a p -adic algebraic number. If $\text{Supp}(f)$ is bounded, then it must be a finite set.

With the help of Theorem 1.7, we prove this conjecture with an additional assumption:

thm:16129

Theorem 1.10 (cf. Corollary 6.24). — Let $f \in \mathbf{L}_p$ be a p -adic algebraic number with bounded support. If $\text{Supp}(f)$ has only finitely many accumulation points, then it must be a finite set.

rmk:37294

Remark 1.11. — By accumulation points of $\text{Supp}(f) \subseteq \mathbf{Q}$ in Theorem 1.10 and other occurrences in this paper, we mean the accumulation point of $\text{Supp}(f)$ in $\mathbf{R}$. In fact, it makes no difference whether one interprets accumulation points in $\mathbf{Q}$ or in $\mathbf{R}$. By [Lam86, Theorem 2], the set of all p -adic Hahn series f that the accumulation points of $\text{Supp}(f)$ are rational numbers is an algebraically closed subfield F of $\mathbf{L}_p$. Since F contains $\check{\mathbf{Q}}_p$, it contains all $\check{\mathbf{Q}}_p$ -algebraic, and

hence all $\mathbf{Q}_p$ -algebraic, p -adic Hahn series. This subtlety is identified during the formalization of this work.

A direct corollary of this theorem is the following:

coro:3344

Corollary 1.12. — Let $q_1 < q_2 < \dots < q_n < \dots$ be a strictly increasing sequence of rational numbers. If a p -adic Hahn series $f = \sum_{i=1}^{\infty} [f(q_i)]p^{q_i} \in \mathbf{L}_p$ is algebraic over $\check{\mathbf{Q}}_p$ (resp. $\mathbf{Q}_p$), then one must have $\lim_{i \rightarrow \infty} q_i = \infty$.

We point out that this corollary actually produces infinitely many counterexamples to the first question in Question 1.2. For example, as is mentioned in [Ked01b, Section 1], a prototype of Kedlaya's criterion for the $\overline{\mathbf{F}}_p$ -algebraicity of Hahn series in $\mathbf{L}_p^b$ is the following result of Huang, which was independently discovered by Ștefănescu:

Proposition 1.13 (cf. [Hua68; Ște83]). — Let $f = \sum_{i=1}^{\infty} f(i) \cdot t^{-1/p^i} \in \mathbf{L}_p^b$ be a Hahn series. Then the following are equivalent:

1. the series f is algebraic over $\mathbf{F}_p((t))$;
2. the series f is algebraic over $\overline{\mathbf{F}}_p((t))$;
3. the sequence $\{f(i)\}_{i \geq 1}$ is eventually periodic.

The p -adic analogue of this result is immediate from Corollary 1.12:

prop:31772

Proposition 1.14. — Let $f = \sum_{i=1}^{\infty} [f(i)] \cdot p^{-1/p^i} \in \mathbf{L}_p$ be a p -adic Hahn series. Then the following are equivalent:

1. the series f is algebraic over $\mathbf{Q}_p$;
2. the series f is algebraic over $\check{\mathbf{Q}}_p$;
3. $f(i) = 0$ for all but finitely many i .

We hope the approach in this article can be further developed to give a full answer to Conjecture 1.9.

1.4. Formalization in Lean. — Considering the highly combinatorial nature of the sparseness condition and the transcendence proof, we formalize all results of this article in Lean 4 with Mathlib v4.31.0, a popular proof assistant based on dependent type theory, with the help of the agentic auto-formalization system Archon (cf. [Ju+26]) developed by AI4Math team at BICMR, Peking University.

Remark 1.15. — The formalization of all contents of Section 2, Section 3, Section 4 and Section 5 is completely **sorry**-free. For the formalization of Section 6 and the consequential Corollary 1.12 and Proposition 1.14, we use two results of Kedlaya (cf. Theorem 6.1 and Theorem 6.4) as black boxes, for they require a significant amount of work to formalize and are far beyond the scope of this article. These are the only two **sorry**-s in the formalization project.

The formalization is available at <https://github.com/YijunYuan/FormalizedSparse>, and we refer the reader to Appendix A for a detailed discussion on the formalization process.

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2. Preliminaries on Hahn series

sec:18373

To keep this article self-contained, we briefly recall some basic facts about Hahn series.

def:61947

Definition 2.1 ([Poo93, Section 3]). — Let R be a commutative ring and G be an ordered group.

1. For any $f \in \text{Hom}_{\text{Set}}(G, R)$, we define the **support** of f to be

$$\text{Supp}(f) = \{g \in G : f(g) \neq 0\}.$$

2. Define the set of **Hahn series** over R with value group G to be

$$R\langle\langle G \rangle\rangle := \{f \in \text{Hom}_{\text{Set}}(G, R) : \text{Supp}(f) \text{ is well-ordered}\}.$$

By introducing a formal variable t , elements in $R\langle\langle G \rangle\rangle$ will also be written as $\sum_{g \in G} r_g t^g$, where $r_g \in R$ for all $g \in G$.

prop:40704

Proposition 2.2 ([Poo93, Lemma 1, Corollary 2]). — Let R be a commutative ring and G be an ordered group.

1. With identity $1 \cdot t^0$ and addition as well as multiplication given by

$$\sum_{g \in G} a_g t^g + \sum_{g \in G} b_g t^g := \sum_{g \in G} (a_g + b_g) t^g, \quad \sum_{g \in G} a_g t^g \cdot \sum_{g \in G} b_g t^g := \sum_{g \in G} \left(\sum_{h \in G} a_h b_{g-h} \right) t^g,$$

$R\langle\langle G \rangle\rangle$ forms a commutative ring.

2. If R is a field, then so is $R\langle\langle G \rangle\rangle$. Moreover, with the map

$$v : R\langle\langle G \rangle\rangle \longrightarrow G \cup \{\infty\}, \quad f \longmapsto \begin{cases} \min \text{Supp}(f), & \text{if } f \neq 0 \\ \infty, & \text{if } f = 0 \end{cases},$$

$R\langle\langle G \rangle\rangle$ becomes a valued field with value group G and residue field R .

Since $\text{char } R\langle\langle G \rangle\rangle = \text{char } R$, we call $R\langle\langle G \rangle\rangle$ the **equal-characteristic field of Hahn series** over R with value group G , also denoted by $R\langle\langle t^G \rangle\rangle$ with respect to the formal variable t .

prop:49318

Proposition 2.3 ([Poo93, Proposition 3, Corollary 3, Proposition 5])

Let k be a perfect field of characteristic p and G be an ordered group containing $\mathbb{Z}$ as a subgroup. Besides that, let

$$\mathcal{N} := \left\{ \sum_{g \in G} r_g t^g \in W(k)\langle\langle t^G \rangle\rangle : \text{for every } g \in G, \sum_{n \in \mathbb{Z}} r_{g+n} p^n = 0 \right\},$$

where $W(k)$ is the ring of Witt vectors of k . We call elements in $\mathcal{N}$ the **null series** of $W(k)\langle\langle t^G \rangle\rangle$. Then

1. $\mathcal{N}$ is a maximal ideal of $W(k)\langle\langle t^G \rangle\rangle$, which makes $W(k)\langle\langle p^G \rangle\rangle := W(k)\langle\langle t^G \rangle\rangle / \mathcal{N}$ a field⁽¹⁾, called the **field of p -adic Hahn series**.
2. Every element x in $W(k)\langle\langle p^G \rangle\rangle$ can be uniquely written as

$$x = \sum_{g \in G} [r_g] p^g,$$

where $r_g \in k$ for all $g \in G$ and $[\cdot] : k \longrightarrow W(k)$ is the Teichmüller lift. We call this the **standard expansion** of the element x .

3. For $f = \sum_{g \in G} [r_g] p^g \in W(k)\langle\langle p^G \rangle\rangle$, define the **support** of f to be

$$\text{Supp}(f) = \{g \in G : r_g \neq 0\}.$$

Then the map

$$v : W(k)\langle\langle p^G \rangle\rangle \longrightarrow G \cup \{\infty\}, \quad f \mapsto \begin{cases} \min \text{Supp}(f), & \text{if } f \neq 0 \\ \infty, & \text{if } f = 0 \end{cases}$$

⁽¹⁾Informally, $W(k)\langle\langle p^G \rangle\rangle$ is obtained by replacing the formal variable t in elements of $W(k)\langle\langle t^G \rangle\rangle$ by the prime p .

makes $W(k)((G))/\mathcal{N}$ a mixed-characteristic valued field with value group G and residue field k .

The most fundamental property of the field of Hahn series is the following:

Theorem 2.4 (cf. [Poo93, Theorem 1, Corollary 4, Corollary 6])

Let F be an equal-characteristic (resp. mixed-characteristic) valued field with divisible value group G and algebraically closed residue field k . Then the equal-characteristic (resp. p -adic) field of Hahn series $k((t^G))$ (resp. $W(k)((p^G))$) is the unique (up to isomorphism of valued fields) minimal spherically complete extension of F . Moreover, it is algebraically closed and complete.

The following are the fields of Hahn series used in this article:

eg:44541

Example 2.5. — Let $F = \overline{\mathbf{F}}_p((t))$ (resp. $\check{\mathbf{Q}}_p$), which has value group $\mathbf{Q}$ and residue field $\overline{\mathbf{F}}_p$. Then the field of equal-characteristic (resp. p -adic) Hahn series $\mathbf{L}_p^b := \overline{\mathbf{F}}_p((t^{\mathbf{Q}}))$ (resp. $\mathbf{L}_p := \check{\mathbf{Z}}_p((p^{\mathbf{Q}})) = W(\overline{\mathbf{F}}_p)((p^{\mathbf{Q}}))$) is the spherical completion of F with the same residue field and value group, and is algebraically closed and complete.

For brevity, we call elements of $\mathbf{L}_p$ p -adic Hahn series without specifying the residue field and value group.

3. Sparseness and (c, n) -sparseness

sec:sparse

To give a rigorous and workable formulation of the sparseness condition in Definition 1.4, we introduce several combinatorial notions and auxiliary functions related to base- p expansions of rational numbers in $[0, 1)$.

Definition 3.1. —

1. For $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, let $\Psi(\underline{d}) := \sum_{i=1}^{\infty} d_i \in \mathbf{N}$ and $\|\underline{d}\| := \sum_{i=1}^{\infty} d_i p^{-i} \in \mathbf{Q}_{\geq 0}$.
2. Let $\mathbb{P} := \bigoplus_{\mathbf{Z}_{\geq 1}} \{0, 1, \dots, p-1\} \subsetneq \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$.
3. For $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$ and $i \in \mathbf{Z}_{\geq 1}$, denote by $\underline{d}_i$ the i -th component of $\underline{d}$.

Lemma 3.2. — For any $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, there exists a unique element $\tau(\underline{d}) \in \mathbb{P}$ such that $\|\underline{d}\| - \|\tau(\underline{d})\| \in \mathbf{Z}$.

Proof. — If one writes $\|\underline{d}\|$ as a decimal expansion⁽²⁾ in base p :

$$\|\underline{d}\| = w.d_1 \cdots d_n \cdots,$$

where $w \in \mathbf{Z}$ and $d_i \in \{0, 1, \dots, p-1\}$ for any i , then $\tau(\underline{d}) = 0.d_1 \cdots d_n \cdots$. The uniqueness is trivial. $\square$

We collect several properties of these concepts in the following lemma:

lem:18813

Lemma 3.3. — Let $\underline{d}, \underline{e} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$.

it:28363

it:52935

1. The maps $\Psi(\cdot)$ and $\|\cdot\|$ are additive. Moreover, $\|\cdot\|$ is injective when restricted to $\mathbb{P}$.
2. One has $\tau(\underline{d}) = \tau(\underline{e})$ if and only if $\|\underline{d}\| - \|\underline{e}\| \in \mathbf{Z}$.
3. $\underline{d} \in \mathbb{P}$ if and only if $\tau(\underline{d}) = \underline{d}$.
4. One has $\Psi(\tau(\underline{d})) \leq \Psi(\underline{d})$, with equality if and only if $\underline{d} \in \mathbb{P}$.

lem:17816

Proof. — The first three statements are straightforward. We only prove the last statement.

For any $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, let

$$N(\underline{d}) := \begin{cases} 0, & \text{if } \underline{d} \in \mathbb{P}; \\ \max\{i | d_i \geq p\}, & \text{otherwise.} \end{cases}$$

⁽²⁾We do not allow infinite strings of $(p-1)$ in the decimal expansion of $\|\underline{d}\|$ to ensure the uniqueness of $\tau(\underline{d})$.

By a descent argument on $N(\underline{d})$, it suffices to show that if $N(\underline{d}) \geq 1$, then there exists $\underline{e} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$ such that $\|\underline{d}\| - \|\underline{e}\| \in \mathbf{Z}$, $\Psi(\underline{e}) < \Psi(\underline{d})$, and $N(\underline{e}) < N(\underline{d})$.

Writing $d_{N(\underline{d})} = p \cdot r + s$ with $r \in \mathbf{N}$ and $s \in \{0, 1, \dots, p-1\}$, we take $\underline{e} := (d_0, \dots, d_{N(\underline{d})-2}, d_{N(\underline{d})-1} + r, s, 0, \dots)$. Then

$$\Psi(\underline{d}) - \Psi(\underline{e}) = d_{N(\underline{d})} - r - s = (p-1) \cdot r > 0$$

and $\|\underline{d}\| - \|\underline{e}\| = 0$. The result follows. $\square$

Remark 3.4. — For a rational number q in $[0, 1)$ with a finite-length decimal expansion in base p , the preimage $\underline{q} \in \mathbb{P}$ of q under the map $\|\cdot\|$ extracts the digits of q in base p , and $\Psi(\underline{q})$ is the p -digit sum of q .

def:38144

Definition 3.5. — Let p be a prime number. Let $c, n \geq 1$ be integers. A subset $S \subset \mathbb{P}$ is (c, n) -sparse if

1. there exists $c \geq 1$ such that $\Psi(\underline{d}) \leq c$ for every $\underline{d} \in S$;
2. there exist n (not necessarily distinct) elements $\underline{d}^{(1)}, \underline{d}^{(2)}, \dots, \underline{d}^{(n)} \in S$ such that
 - (a) $\Psi(\underline{d}^{(i)}) = c$ for $i = 1, 2, \dots, n$ and $\sum_{j=1}^n \underline{d}_j^{(i)} < p$ for $j \in \mathbf{N}$ (that is, $\sum_{i=1}^n \underline{d}^{(i)} \in \mathbb{P}$);
 - (b) if $\underline{e}^{(1)}, \underline{e}^{(2)}, \dots, \underline{e}^{(n)} \in S$ satisfy $\left\| \sum_{i=1}^n \underline{d}^{(i)} \right\| - \left\| \sum_{i=1}^n \underline{e}^{(i)} \right\| \in \mathbf{Z}$, then up to a permutation of $\{1, 2, \dots, n\}$, $\underline{d}^{(i)} = \underline{e}^{(i)}$ for every $i = 1, 2, \dots, n$.

The following lemma reformulates the sparseness condition in Definition 1.4 in terms of the (c, n) -sparseness condition in Definition 3.5.

Lem:54409

Lemma 3.6. — A subset $W \neq \{0\}$ of $[0, 1) \cap \mathbf{Q}$ is sparse in the sense of Definition 1.4 if and only if there exists a subset S of $\mathbb{P}$ such that $W = \|S\|$ and there exists an integer $c \geq 1$ such that S is (c, n) -sparse for infinitely many integers $n \geq 1$.

Proof. — Suppose that $W \subset [0, 1) \cap \mathbf{Q}$ is sparse. Then the p -digit sum of every element in W is finite and bounded by $\text{dom}_p(W)$. Thus we may take

$$S := \left\{ (q_i)_{i \in \mathbf{Z}_{\geq 1}} \mid q = \sum_{i=0}^{\infty} q_i p^{-i} \in W, q_i \in \{0, \dots, p-1\} \right\}.$$

Then S is $(\text{dom}_p(W), n)$ -sparse for infinitely many integers $n \geq 1$.

The converse direction follows by a similar argument. $\square$

We give a concrete example of a sparse set, which is the prototype of the situation in which the sparseness condition is satisfied:

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Example 3.7. — Let $\underline{A} := (A_i)_{i \geq 1}$ be a family of pairwise disjoint nonempty subsets of $\mathbf{Z}_{\geq 1}$ such that $\sup_i |A_i| < \infty$ and this supremum is attained by infinitely many i . Then the set

$$M(\underline{A}) := \left\{ \sum_{r \in A_i} p^{-r} \mid i = 1, 2, \dots \right\} \subset [0, 1) \cap \mathbf{Q}$$

is sparse.

Proof. — For any i , one has $\mathfrak{N}_p(\sum_{r \in A_i} p^{-r}) = |A_i|$, and consequently $\text{dom}_p(M(\underline{A})) = \sup_i |A_i| < \infty$. Since the supremum is attained by infinitely many i , the set $\text{Dom}_p(M(\underline{A}))$ is infinite:

$$\text{Dom}_p(M(\underline{A})) = \left\{ \mathbf{d}_j := \sum_{r \in A_{i_j}} p^{-r} \mid j = 1, 2, \dots \right\},$$

with $\mathbf{d}_k \neq \mathbf{d}_l$ for any $k \neq l$.

Take $e_1, \dots, e_n \in M(\underline{A})$ such that $\sum_{j=1}^n e_j - \sum_{j=1}^n \mathbf{d}_j \in \mathbf{Z}$. Since $M(\underline{A}) \subset \|\mathbb{P}\|$, we may write $e_j = \|\underline{e}^{(j)}\|$ and $\mathbf{d}_j = \|\underline{\mathbf{d}}^{(j)}\|$ with $\underline{e}^{(j)}, \underline{\mathbf{d}}^{(j)} \in \mathbb{P}$ for every $j = 1, \dots, n$. Then one has $\sum_{j=1}^n \mathbf{d}_j = \sum_{r \in \bigsqcup_{j=1}^n A_{i_j}} p^{-r}$, and consequently $\sum_{j=1}^n \underline{\mathbf{d}}^{(j)} \in \mathbb{P}$. In particular,

$$n \cdot \text{dom}_p(M(\underline{A})) = \mathfrak{N}_p \left(\sum_{r \in \bigsqcup_{j=1}^n A_{i_j}} p^{-r} \right) = \mathfrak{N}_p \left(\sum_{j=1}^n \mathbf{d}_j \right) = \Psi \left(\sum_{j=1}^n \underline{\mathbf{d}}^{(j)} \right).$$

Since

$$\begin{aligned} \Psi \left(\sum_{j=1}^n \underline{\mathbf{d}}^{(j)} \right) &= \Psi \left(\tau \left(\sum_{j=1}^n \underline{\mathbf{d}}^{(j)} \right) \right) = \Psi \left(\tau \left(\sum_{j=1}^n \underline{e}^{(j)} \right) \right) \\ &\leq \Psi \left(\sum_{j=1}^n \underline{e}^{(j)} \right) = \sum_{j=1}^n \Psi(\underline{e}^{(j)}) \leq n \cdot \text{dom}_p(M(\underline{A})), \end{aligned}$$

one concludes that $\Psi(\underline{e}^{(j)}) = \text{dom}_p(M(\underline{A}))$ for every $j = 1, \dots, n$ and $\sum_{j=1}^n \underline{e}^{(j)} \in \mathbb{P}$. Consequently, one has $\sum_{j=1}^n \underline{e}^{(j)} = \sum_{j=1}^n \underline{\mathbf{d}}^{(j)}$, implying that $\sum_{j=1}^n e_j = \sum_{j=1}^n \mathbf{d}_j$.

Note that there is no duplication among $e_1, \dots, e_n$: if not, then there exists a coordinate of $\sum_{j=1}^n \underline{e}^{(j)}$ that is at least 2, contradicting the fact that every coordinate of $\sum_{j=1}^n \underline{\mathbf{d}}^{(j)}$ is 0 or 1. If one writes $e_j = \sum_{r \in A_{k_j}} p^{-r}$ for every $j = 1, \dots, n$, then

$$\sum_{r \in \bigsqcup_{j=1}^n A_{k_j}} p^{-r} = \sum_{j=1}^n e_j = \sum_{j=1}^n \mathbf{d}_j = \sum_{r \in \bigsqcup_{j=1}^n A_{i_j}} p^{-r},$$

implying that $\bigsqcup_{j=1}^n A_{k_j} = \bigsqcup_{j=1}^n A_{i_j}$. This forces $A_{k_j} = A_{i_j}$ and consequently $e_j = \mathbf{d}_j$ for every $j = 1, \dots, n$, up to a permutation of $\{1, \dots, n\}$. $\square$

We end this section with the following technical lemma, which will be used in the proof of Theorem 1.7 to extract the orphaned exponents in the multinomial expansion.

Lem:39609

Lemma 3.8. — *Let $S \subset \mathbb{P}$ be a (c, n) -sparse subset for some integers $c, n \geq 1$. Let $\underline{d}^{(1)}, \dots, \underline{d}^{(n)}$ be elements of S that satisfy the condition (2) of Definition 3.5 and set*

$$\phi_0: S \longrightarrow \mathbf{N}, \quad \underline{d} \longmapsto \text{card} \left(\left\{ i \mid 1 \leq i \leq n, \underline{d} = \underline{d}^{(i)} \right\} \right).$$

Then ϕ_0 is the unique function $\phi: S \rightarrow \mathbf{N}$ such that $\sum_{\underline{d} \in S} \phi(\underline{d}) \leq n$ and

$$\sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi(\underline{d}) \equiv \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) \pmod{\mathbf{Z}}.$$

Proof. — Suppose that $\phi_1: S \rightarrow \mathbf{N}$ is another function such that $\sum_{\underline{d} \in S} \phi_1(\underline{d}) \leq n$ and

$$\sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_1(\underline{d}) \equiv \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) \pmod{\mathbf{Z}}.$$

1. If $\sum_{\underline{d} \in S} \phi_1(\underline{d}) < n$, then by Lemma 3.3 (4) we have

$$\Psi \left(\tau \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_1(\underline{d}) \right) \right) \leq \Psi \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_1(\underline{d}) \right) = \sum_{\underline{d} \in S} \Psi(\underline{d}) \cdot \phi_1(\underline{d}) < c \cdot n.$$

On the other hand, the sparseness condition (2a) of Definition 3.5 implies that $\sum_{j=1}^n \underline{d}_i^{(j)} < p$ for any i , hence

$$\sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) = \left(\sum_{j=1}^n \underline{d}_i^{(j)} \right)_{i \in \mathbf{Z}_{\geq 0}} = \tau \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) \right)$$

and consequently

$$\Psi \left(\tau \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) \right) \right) = \sum_{i=0}^{\infty} \left(\sum_{j=1}^n \underline{d}_i^{(j)} \right) = \sum_{j=1}^n \Psi(\underline{d}^{(j)}) = n \cdot c.$$

Thus, we have $\tau \left(\sum_{\underline{d} \in S} \phi_1(\underline{d}) \cdot \underline{d} \right) \neq \tau \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) \right)$, which leads to a contradiction by Lemma 3.3 (2).

2. If $\sum_{\underline{d} \in S} \phi_1(\underline{d}) = n$, then we take $\underline{e}^{(1)}, \underline{e}^{(2)}, \dots, \underline{e}^{(n)} \in S$ such that $\phi_1(\underline{d}) = \text{card}(\{i | 1 \leq i \leq n, \underline{d} = \underline{e}^{(i)}\})$ for any $\underline{d} \in S$. Then

$$\sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_1(\underline{d}) - \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) = \left\| \sum_{\underline{d} \in S} \underline{d} \cdot \phi_1(\underline{d}) \right\| - \left\| \sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) \right\| = \left\| \sum_{i=1}^n \underline{e}^{(i)} \right\| - \left\| \sum_{i=1}^n \underline{d}^{(i)} \right\| \in \mathbf{Z}.$$

By the sparseness condition (2b) of Definition 3.5, up to a permutation of $\{1, 2, \dots, n\}$, $\underline{d}^{(i)} = \underline{e}^{(i)}$ for any $i = 1, 2, \dots, n$. This implies that $\phi_1 = \phi_0$. □

4. T -scaled realization of $\mathbf{L}_p$

sec:tscaled

To prove Theorem 1.7, one has to group the terms of a p -adic Hahn series $f = \sum_{q \in \mathbf{Q}} [f(q)]p^q$ in $\mathbf{L}_p$ by the residue of the exponent modulo $\frac{1}{T}\mathbf{Z}$ for some integer $T \geq 1$. This works when $T = 1$, since a direct computation shows that the element

$$\sum_{q \in \text{Supp}(f)/\mathbf{Z}} \left(\sum_{w \in \mathbf{Z}} f(q+w)p^w \right) t^q \in \check{\mathbf{Z}}_p((t^{\mathbf{Q}}))$$

is a preimage of f under the natural projection $\check{\mathbf{Z}}_p((t^{\mathbf{Q}})) \rightarrow \mathbf{L}_p$, where

$$\text{Supp}(f)/\mathbf{Z} := \{\inf(\text{Supp}(f) \cap (q + \mathbf{Z})) | q \in \text{Supp}(f)\} \subset \mathbf{Q}$$

is a well-ordered set of representatives of $\text{Supp}(f)$ modulo $\mathbf{Z}$. However, when $T > 1$, the same construction fails, because the element $\sum_{w \in \frac{1}{T}\mathbf{Z}} f(q+w)p^w$ does not necessarily lie in $\check{\mathbf{Z}}_p$, so a direct analogue of the above construction is not well-defined⁽³⁾. To resolve this, we enlarge the ring $\check{\mathbf{Z}}_p = W(\bar{\mathbf{F}}_p)$ to include $p^{1/T}$, and realize $\mathbf{L}_p$ as a quotient of $\check{\mathbf{Z}}_p[p^{1/T}](t^{\mathbf{Q}})$ by a suitable ideal. This is the main content of this section.

From now on, T will be a positive integer. We set $\check{\mathbf{Z}}_{p,T} := \check{\mathbf{Z}}_p[p^{1/T}]$ and $\check{\mathbf{Q}}_{p,T} := \check{\mathbf{Q}}_p(p^{1/T})$.

Lem:61648

Lemma 4.1. — One has $[\check{\mathbf{Q}}_{p,T} : \check{\mathbf{Q}}_p] = T$. In particular, $1, p^{1/T}, \dots, p^{(T-1)/T}$ form a basis of $\check{\mathbf{Q}}_{p,T}$ over $\check{\mathbf{Q}}_p$.

Proof. — Since $\check{\mathbf{Q}}_p$ is a discrete valuation field and $p^{1/T}$ is a root of the Eisenstein polynomial $X^T - p$, the result follows from the Eisenstein criterion. □

⁽³⁾This issue was detected during the formalization of the proof of the main theorem in Lean 4.

Remark 4.2. — $\check{\mathbf{Z}}_p((t^{\mathbf{Q}}))$ is a subfield of $\check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))$ via the natural inclusion $\check{\mathbf{Z}}_p \subseteq \check{\mathbf{Z}}_{p,T}$.

Definition 4.3. — Let $\mathcal{N}_T$ be the set of elements $\sum_{q \in \mathbf{Q}} c_q t^q \in \check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))$ such that for any $q \in \mathbf{Q}$,

$$\sum_{n \in \mathbf{Z}} c_{q + \frac{n}{T}} p^{\frac{n}{T}} = 0.$$

We call these elements the *T-null-series* in $\check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))$.

Remark 4.4. — When $T = 1$, $\mathcal{N}_T$ coincides with the ideal $\mathcal{N}$ of null-series in $\check{\mathbf{Z}}_p((t^{\mathbf{Q}}))$ defined in Proposition 2.3.

Lemma 4.5. —

1. $\mathcal{N}_T$ is an ideal of $\check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))$.
2. For every element f of $\check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))$, there exists a unique element $g = \sum_{q \in \mathbf{Q}} [g(q)] t^q$ in $\check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))$ such that $f - g \in \mathcal{N}_T$.
3. $\mathcal{N}_T$ is a maximal ideal of $\check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))$, so that $\check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))/\mathcal{N}_T$ is a field.

Proof. — These three statements are generalizations of [Poo93, Proposition 3], [Poo93, Proposition 4] and [Poo93, Corollary 3] respectively, and the proofs are essentially the same. $\square$

Remark 4.6. — By (2) of this lemma, we will formally write elements of $\check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))/\mathcal{N}_T$ as $\sum_{q \in \mathbf{Q}} [g(q)] p^q$.

Lem:14623

Lemma 4.7. — One has $\mathcal{N}_T \cap \check{\mathbf{Z}}_p((t^{\mathbf{Q}})) = \mathcal{N}$.

Proof. — Let $f = \sum_{q \in \mathbf{Q}} [f(q)] t^q \in \mathcal{N}$, then for any $q \in \mathbf{Q}$,

eq:64661

$$(a) \quad \sum_{n \in \mathbf{Z}} [f(q + n)] p^n = 0.$$

Notice that for any $q \in \mathbf{Q}$, one has

$$\begin{aligned} \sum_{n \in \mathbf{Z}} \left[f\left(q + \frac{n}{T}\right) \right] p^{\frac{n}{T}} &= \sum_{u=0}^{T-1} \sum_{\substack{n \in \mathbf{Z} \\ n \equiv u \pmod{T}}} \left[f\left(q + \frac{n}{T}\right) \right] p^{\frac{n}{T}} \\ &= \sum_{u=0}^{T-1} p^{\frac{u}{T}} \sum_{n \in \mathbf{Z}} \left[f\left(\left(q + \frac{u}{T}\right) + n\right) \right] p^n. \end{aligned}$$

By applying (a) to $q + \frac{u}{T}$ for $u = 0, 1, \dots, T-1$, one has $\sum_{n \in \mathbf{Z}} [f(q + \frac{n}{T})] p^{\frac{n}{T}} = 0$. As a result, $f \in \mathcal{N}_T$.

Conversely, let $g = \sum_{q \in \mathbf{Q}} [g(q)] t^q \in \mathcal{N}_T$. Then for any $q \in \mathbf{Q}$, one has

$$\sum_{n \in \mathbf{Z}} \left[g\left(q + \frac{n}{T}\right) \right] p^{\frac{n}{T}} = \sum_{u=0}^{T-1} p^{\frac{u}{T}} \sum_{n \in \mathbf{Z}} \left[g\left(\left(q + \frac{u}{T}\right) + n\right) \right] p^n = 0.$$

By Lemma 4.1, for any $u = 0, 1, \dots, T-1$, one has $\sum_{n \in \mathbf{Z}} [g((q + \frac{u}{T}) + n)] p^n = 0$. The result follows from taking $u = 0$. $\square$

Lem:21624

Lemma 4.8. — One has $\check{\mathbf{Z}}_p((t^{\mathbf{Q}})) + \mathcal{N}_T = \check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))$.

Proof. — Take $f = \sum_{q \in \mathbf{Q}} c_q t^q \in \check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))$, where

$$c_q = \sum_{n=0}^{\infty} [c_{q,n}] p^{\frac{n}{T}} \in \check{\mathbf{Z}}_{p,T}.$$

Then one has $f = f_0 + p^{\frac{1}{T}} f_1 + \cdots + p^{\frac{T-1}{T}} f_{T-1}$, where

$$f_u = \sum_{t \in \mathbf{Q}} \left(\sum_{\substack{n \in \mathbf{N} \\ n \equiv u \pmod{T}}} [c_{q,n}] p^{\frac{n-u}{T}} \right) t^q \in \check{\mathbf{Z}}_p((t^{\mathbf{Q}})), \quad u = 0, 1, \dots, T-1.$$

It suffices to show that $p^{\frac{u}{T}} f_u$ belongs to $\check{\mathbf{Z}}_p((t^{\mathbf{Q}})) + \mathcal{N}_T$ for $u = 0, 1, \dots, T-1$. This is clear if we write f_u as $f_u = (p^{\frac{u}{T}} \cdot t^0 - t^{\frac{u}{T}}) f_u + t^{\frac{u}{T}} f_u$, where $t^{\frac{u}{T}} f_u \in \check{\mathbf{Z}}_p((t^{\mathbf{Q}}))$ and $(p^{\frac{u}{T}} \cdot t^0 - t^{\frac{u}{T}}) f_u$ is a T -null-series. $\square$

prop:18592

Proposition 4.9. — *There is an isomorphism of valued fields*

$$\sigma: \mathbf{L}_p \xrightarrow{\cong} \check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))/\mathcal{N}_T.$$

Moreover, the following diagram commutes:

$$(b) \quad \begin{array}{ccc} \check{\mathbf{Z}}_p((t^{\mathbf{Q}})) & \xhookrightarrow{\iota} & \check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}})) \\ \downarrow \pi & & \downarrow \pi_T \\ \mathbf{L}_p & \xrightarrow[\sigma]{\cong} & \check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))/\mathcal{N}_T \end{array}.$$

eq:28959

In particular, σ sends an element $\sum_{q \in \mathbf{Q}} [f(q)] p^q$ in $\mathbf{L}_p$ to the element $\sum_{q \in \mathbf{Q}} [f(q)] p^q$ in $\check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))/\mathcal{N}_T$.

Proof. — This is a direct consequence of the following standard fact in commutative algebra: given domains $R_1 \subset R_2$ and an ideal I of R_2 , if $R_1 + I = R_2$, then $R_1/(I \cap R_1) \cong R_2/I$, where the isomorphism is induced by the inclusion $R_1 \subset R_2$. $\square$

As an application, the T -scaled realization in Proposition 4.9 gives the desired grouping of terms of a p -adic Hahn series by residue modulo $\frac{1}{T} \mathbf{Z}$ in the case $T > 1$, and provides an explicit lift of $\sigma(f)$ that will serve as the starting point for the proof of the main theorem. We record this lift below.

Let $f = \sum_{q \in \mathbf{Q}} [f(q)] p^q \in \mathbf{L}_p$ and let $W \subset [0, 1) \cap \mathbf{Q}$ be a set of representatives of $-T \cdot \text{Supp}(f)$ modulo $\mathbf{Z}$. Intuitively, the field $\check{\mathbf{Z}}_{p,T}((t^{\mathbf{Q}}))/\mathcal{N}_T$ provides a suitable setting to write f as $\sum_{q \in W} C_{-\frac{q}{T}} p^{-\frac{q}{T}}$, where $C_{-\frac{q}{T}} \in \check{\mathbf{Z}}_{p,T}$ for every $q \in W$. To be more precise, for every $q \in W$, we set

$$n_q := -\frac{q}{T} - \inf \left(\text{Supp}(f) \cap \left(-\frac{q}{T} + \frac{1}{T} \mathbf{Z} \right) \right) \in \frac{1}{T} \mathbf{Z}.$$

This element is well-defined. Indeed, the set $\text{Supp}(f) \cap \left(-\frac{q}{T} + \frac{1}{T} \mathbf{Z} \right)$ is nonempty by the assumption that $-\frac{1}{T} W$ is a set of representatives of $\text{Supp}(f)$ modulo $\frac{1}{T} \mathbf{Z}$. Moreover, it is a subset of the well-ordered set $\text{Supp}(f)$, and hence has a minimum element. Then the set

$$\widetilde{W} := \left\{ -\frac{q}{T} - n_q \mid q \in W \right\} \subseteq \text{Supp}(f)$$

is a well-ordered set of representatives of $\text{Supp}(f)$ modulo $\frac{1}{T} \mathbf{Z}$, and is in bijection with W via the map

$$(c) \quad \mu_W: W \longrightarrow \widetilde{W}, \quad q \longmapsto -\frac{q}{T} - n_q = \inf \left(\text{Supp}(f) \cap \left(-\frac{q}{T} + \frac{1}{T} \mathbf{Z} \right) \right).$$

For any $w \in \widetilde{W}$, set $C_w := \sum_{r \in \mathbf{Z}} [f(w + \frac{r}{T})] p^{\frac{r}{T}} \in \check{\mathbf{Q}}_{p,T}$. By the construction of $\widetilde{W}$, we have $C_w \neq 0$ and $f(w + \frac{r}{T}) = 0$ for every $w \in \widetilde{W}$ and every integer $r < 0$. Thus $C_w \in \check{\mathbf{Z}}_{p,T}$, and the

eq:mu-bijection

element

eq:20385

$$(d) \quad \widehat{f} := \sum_{w \in \widetilde{W}} C_w \cdot t^w \in \check{\mathbf{Z}}_{p,T}(\langle t^{\mathbf{Q}} \rangle)$$

is a lift of $\sigma(f)$ in $\check{\mathbf{Z}}_{p,T}(\langle t^{\mathbf{Q}} \rangle)$.

sec:main

5. Main theorem

In this section, we prove the main theorem of this paper (cf. Theorem 5.3): a p -adic Hahn series whose sign-inverted support, after scaling by some integer $T \geq 1$, admits a sparse set of representatives modulo $\mathbf{Z}$ is transcendental over $\check{\mathbf{Q}}_p$. The argument proceeds by contradiction: if such a Hahn series f is a root of a polynomial P over $\check{\mathbf{Q}}_p$, then expanding the lifted element $P(\widehat{f})$ from (d) via the multinomial formula would yield a family of vanishing identities on its coefficients (Lemma 5.1). The sparseness condition then forces these identities, when specialized at a carefully chosen exponent, to collapse to a single nonzero term (Lemma 5.2), leading to a contradiction.

We isolate the two key inputs as separate lemmas before assembling the proof.

-from-polynomial

Lemma 5.1. — *Let $f \in \mathbf{L}_p$ be a p -adic Hahn series and let $T \geq 1$ be an integer. Let $W \subset [0, 1) \cap \mathbf{Q}$ be a set of representatives of $-T \cdot \text{Supp}(f)$ modulo $\mathbf{Z}$, and let $\widetilde{W} \subseteq \text{Supp}(f)$ and $\widehat{f} = \sum_{w \in \widetilde{W}} C_w \cdot t^w \in \check{\mathbf{Z}}_{p,T}(\langle t^{\mathbf{Q}} \rangle)$ be as in (d). Suppose that f is a root of a polynomial $P(X) = \sum_{i=0}^n a_i X^i \in \check{\mathbf{Z}}_p[X]$. Then for every $q \in \mathbf{Q}$, one has*

eq:7609

$$(e) \quad \sum_{w \in \mathbf{Z}} p^{\frac{w}{T}} \sum'_{\substack{\widetilde{\phi}: \widetilde{W} \rightarrow \mathbf{N} \\ \sum_{s \in \widetilde{W}} \widetilde{\phi}(s) \leq n \\ \sum_{s \in \widetilde{W}} \widetilde{\phi}(s) \cdot s = q + \frac{w}{T}}} a_{\sum_{s \in \widetilde{W}} \widetilde{\phi}(s)} \frac{(\sum_{s \in \widetilde{W}}'' \widetilde{\phi}(s))!}{\prod_{s \in \widetilde{W}}'' \widetilde{\phi}(s)!} \prod_{s \in \widetilde{W}}'' C_s^{\widetilde{\phi}(s)} = 0,$$

where $\sum'$ emphasizes that the summation is taken over a finite index set, and $\sum''$ (resp. $\prod''$) emphasizes that the function inside the summation (resp. product) is finitely supported.

Proof. — By multinomial expansion, for any integer $i \geq 0$, one has

$$\widehat{f}^i = \sum_{q \in \mathbf{Q}} \left(\sum'_{\substack{s_1, \dots, s_i \in \widetilde{W} \\ s_1 + \dots + s_i = q}} \prod_{k=1}^i C_{s_k} \right) t^q = \sum_{q \in \mathbf{Q}} \left(\sum'_{\substack{\widetilde{\phi}: \widetilde{W} \rightarrow \mathbf{N} \\ \sum_{s \in \widetilde{W}} \widetilde{\phi}(s) = i \\ \sum_{s \in \widetilde{W}} \widetilde{\phi}(s) \cdot s = q}} \frac{i!}{\prod_{s \in \widetilde{W}}'' \widetilde{\phi}(s)!} \prod_{s \in \widetilde{W}}'' C_s^{\widetilde{\phi}(s)} \right) t^q.$$

Since $\sigma(\sum_{i=0}^n a_i f^i) = \sum_{i=0}^n a_i \sigma(f)^i = 0$ and $\widehat{f}$ is a lift of $\sigma(f)$, the element

$$\begin{aligned} \sum_{i=0}^n a_i \widehat{f}^i &= \sum_{q \in \mathbf{Q}} \left(\sum_{i=0}^n a_i \sum'_{\substack{\widetilde{\phi}: \widetilde{W} \rightarrow \mathbf{N} \\ \sum_{s \in \widetilde{W}}'' \widetilde{\phi}(s) = i \\ \sum_{s \in \widetilde{W}}'' \widetilde{\phi}(s) \cdot s = q}} \frac{i!}{\prod_{s \in \widetilde{W}}'' \widetilde{\phi}(s)!} \prod_{s \in \widetilde{W}}'' C_s^{\widetilde{\phi}(s)} \right) t^q \\ &= \sum_{q \in \mathbf{Q}} \left(\sum'_{\substack{\widetilde{\phi}: \widetilde{W} \rightarrow \mathbf{N} \\ \sum_{s \in \widetilde{W}}'' \widetilde{\phi}(s) \leq n \\ \sum_{s \in \widetilde{W}}'' \widetilde{\phi}(s) \cdot s = q}} a_{\sum_{s \in \widetilde{W}}'' \widetilde{\phi}(s)} \frac{(\sum_{s \in \widetilde{W}}'' \widetilde{\phi}(s))!}{\prod_{s \in \widetilde{W}}'' \widetilde{\phi}(s)!} \prod_{s \in \widetilde{W}}'' C_s^{\widetilde{\phi}(s)} \right) t^q \end{aligned}$$

is a T -null-series in $\check{\mathbf{Z}}_{p,T}(\langle t^{\mathbf{Q}} \rangle)$. The equation (e) follows from the definition of T -null-series. $\square$

unique-phi-at-q0

Lemma 5.2. — Let $f \in \mathbf{L}_p$ be a p -adic Hahn series. Let $S \subset \mathbb{P}$ be a (c, n) -sparse subset for some integers $c, n \geq 1$, and let $\underline{d}^{(1)}, \dots, \underline{d}^{(n)} \in S$ satisfy condition (2) of Definition 3.5. Set $\phi_0: S \rightarrow \mathbf{N}$, $\underline{d} \mapsto \text{card}\left(\left\{i \mid \underline{d} = \underline{d}^{(i)}\right\}\right)$. Suppose that $\|S\|$ is a set of representatives of $-T \cdot \text{Supp}(f)$ modulo $\mathbf{Z}$, and let $\widetilde{S} := \|\widetilde{S}\|$ and $\mu := \mu_{\|S\|} \circ \|\cdot\|: S \rightarrow \widetilde{S}$ be as in (c). Set

$$q_0 := -\frac{1}{T} \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}).$$

Then $\widetilde{\phi} = \phi_0 \circ \mu^{-1}$ is the unique function $\widetilde{\phi}: \widetilde{S} \rightarrow \mathbf{N}$ satisfying

$$\sum_{s \in \widetilde{S}}'' \widetilde{\phi}(s) \leq n \quad \text{and} \quad \sum_{s \in \widetilde{S}}'' \widetilde{\phi}(s) \cdot s \equiv q_0 \pmod{\frac{1}{T}\mathbf{Z}}.$$

Proof. — Take $\widetilde{\phi}: \widetilde{S} \rightarrow \mathbf{N}$ that satisfies $\sum_{s \in \widetilde{S}}'' \widetilde{\phi}(s) \leq n$ and $\sum_{s \in \widetilde{S}}'' \widetilde{\phi}(s) \cdot s \equiv q_0 \pmod{\frac{1}{T}\mathbf{Z}}$, and set $\phi := \widetilde{\phi} \circ \mu$. Then

$$\sum_{\underline{d} \in S}'' \phi(\underline{d}) = \sum_{s \in \widetilde{S}}'' \widetilde{\phi}(s) \leq n$$

and

$$\begin{aligned} \sum_{\underline{d} \in S}'' \phi(\underline{d}) \cdot \|\underline{d}\| &= -T \sum_{s \in \widetilde{S}}'' \widetilde{\phi}(s) \cdot s - \sum_{\underline{d} \in S}'' \widetilde{\phi}(\mu(\underline{d})) (T \cdot n_{\|\underline{d}\|}) \\ &\equiv -T \sum_{s \in \widetilde{S}}'' \widetilde{\phi}(s) \cdot s \\ &\equiv -T \cdot q_0 = \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) \pmod{\mathbf{Z}}. \end{aligned}$$

By Lemma 3.8, one knows that $\phi = \phi_0$, i.e. $\widetilde{\phi} = \phi_0 \circ \mu^{-1}$. $\square$

The main theorem now follows from the above two lemmas and the construction of $\widehat{f}$ in (d).

thm:23791

Theorem 5.3. — Let $f \in \mathbf{L}_p$ be a p -adic Hahn series such that there exists an integer $T \geq 1$ for which $-T \cdot \text{Supp}(f)$ admits a sparse set $W \neq \{0\}$ of representatives modulo $\mathbf{Z}$. Then f is transcendental over $\bar{\mathbf{Q}}_p$, and hence over $\mathbf{Q}_p$.

Proof. — Suppose for contradiction that f is a root of a polynomial $P(X) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_0 \in \bar{\mathbf{Z}}_p[X]$ with $a_n \neq 0$. By multiplying $P(X)$ by a suitable power of X , we may assume that there exists $S \subset \mathbb{P}$ such that $\|S\| := \{\|\underline{d}\| \in \mathbf{Q} \mid \underline{d} \in S\}$ is a set of representatives of $-T \cdot \text{Supp}(f)$ modulo $\mathbf{Z}$, and S is (c, n) -sparse for some integer $c \geq 1$ (cf. Lemma 3.6), where n equals the degree of $P(X)$. Let $\underline{d}^{(1)}, \dots, \underline{d}^{(n)} \in S$ be elements satisfying condition (2) of Definition 3.5 for n and c , set

$$\phi_0: S \longrightarrow \mathbf{N}, \quad \underline{d} \longmapsto \text{card}\left(\left\{i \mid 1 \leq i \leq n, \underline{d} = \underline{d}^{(i)}\right\}\right),$$

and let $\tilde{S} := \widetilde{\|S\|}$ and $\mu: S \rightarrow \tilde{S}$ be as in (c) (with $W = \|S\|$).

By Lemma 5.1 applied to $W = \|S\|$, for every $q \in \mathbf{Q}$ one has the identity (e). Specializing to

$$q = q_0 := -\frac{1}{T} \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}),$$

Lemma 5.2 tells us that the only $\tilde{\phi}: \tilde{S} \rightarrow \mathbf{N}$ contributing to the interior summation of (e) at $q = q_0$ is $\tilde{\phi} = \phi_0 \circ \mu^{-1}$. Hence (e) at $q = q_0$ reduces to the single term

$$\frac{p^{\frac{w}{T}} \cdot n!}{\prod_{\underline{d} \in S}'' \phi_0(\underline{d})!} \cdot a_n \cdot \prod_{\underline{d} \in S}'' C_{\mu(\underline{d})}^{\phi_0(\underline{d})} = 0,$$

where $w = T \left(\sum_{s \in \tilde{S}}'' \phi_0(\mu^{-1}(s)) \cdot s - q_0 \right) \in \mathbf{Z}$. This forces $a_n = 0$ or $C_{\mu(\underline{d})} = 0$ for some $\underline{d} \in S$, contradicting $a_n \neq 0$ and $C_s \neq 0$ for every $s \in \tilde{S}$. $\square$

6. Application: p -adic Hahn series with bounded support

sec:55224

The goal of this section is to prove Theorem 1.10 (restated as Corollary 6.24 below): a p -adic Hahn series that is algebraic over $\mathbf{Q}_p$ and whose support is bounded with only finitely many accumulation points must have finite support. The argument proceeds in three steps. First, building on results of Kedlaya, we show that $\mathbf{Q}_p$ -algebraicity forces the coefficient function to be *quasi-twist-recurrent* (QTR), a combinatorial recurrence condition (Proposition 6.6). Second, we analyze the structure of bounded QTR sets and prove that, when the number of accumulation points is finite, such a set decomposes into a finite set together with finitely many pairwise disjoint *rays* (Corollary 6.20). Finally, we show that this ray decomposition yields a sparse set of representatives for the sign-inverted and suitably scaled support (Lemmas 6.21 and 6.22), so that our main transcendence theorem (Theorem 5.3) forces the series to be transcendental unless its support is finite.

In this section, we will write

$$0.q_1 \cdots q_n \cdots := \sum_{i=1}^{\infty} q_i \cdot p^{-i}, \quad q_i \in \{0, 1, \dots, p-1\}$$

to represent the base p expansion of any rational number in $[0, 1)$. We will require that $q_j \neq p-1$ for infinitely many j . In fact, every such expansion occurring in this section has finite length, that is, $q_i = 0$ for $i \gg 1$.

In addition, for any Hahn series $f = \sum_{q \in \mathbf{Q}} f(q) t^q \in \mathbf{L}_p^b$ (resp. $f = \sum_{q \in \mathbf{Q}} [f(q)] p^q \in \mathbf{L}_p$), we will write F_f to denote the function $q \mapsto f(q)$ from $\mathbf{Q}$ to $\bar{\mathbf{F}}_p$.

sec:5330

6.1. Quasi-twist-recurrent functions. — In [Ked01b], Kedlaya characterizes the algebraic closure of $K((t))$ in $K((t^{\mathbf{Q}}))$ for an arbitrary algebraically closed field K of characteristic $p > 0$ in terms of twist-recurrent sequences. In particular, when $K = \overline{\mathbf{F}}_p$, this characterization admits a simpler description⁽⁴⁾:

thm:28713

Theorem 6.1 (cf. [Ked01b, Theorem 15][Ked17, Theorem 11.11])

For $a \in \mathbf{Z}_{>0}$, $b \in \mathbf{Z}$ and $c \in \mathbf{Z}_{\geq 0}$, we define

$$S_{a,b,c} = \left\{ \frac{1}{a} \left(n - \sum_{k=1}^{\infty} b_k p^{-k} \right) \middle| n \geq -b, b_k \in \mathbf{N}_{<p}, \sum b_k \leq c \right\} \subset \frac{1}{a} \mathbf{Z}[1/p]$$

and

$$T_c := S_{1,0,c} \cap (-1, 0) = \left\{ - \sum_{k=1}^{\infty} b_k p^{-k} \middle| b_k \in \mathbf{N}_{<p}, \sum b_k \leq c \right\} \subset \mathbf{Z}[1/p].$$

A Hahn series $x = \sum_q x_q t^q \in \overline{\mathbf{F}}_p((t^{\mathbf{Q}}))$ is integral over $\overline{\mathbf{F}}_p((t))$ if and only if the following conditions hold:

it:23822

1. There exist a, b, c such that $\text{Supp}(x) \subseteq S_{a,b,c}$.
2. For some (hence any) a, b, c as in (1), there exist integers M, N with the following property: for each integer $m \geq -b$, the function $f_m: T_c \rightarrow K$ given by $f_m(z) = x_{(m+z)/a}$ has the property that every sequence of the following form becomes periodic of period N after at most M terms:

$$c_n = f_m \left(- \sum_{k=1}^{j-1} b_k p^{-k} - p^{-n} \left(\sum_{k=j}^{\infty} b_k p^{-k} \right) \right), \quad n = 0, 1, \dots,$$

where $j \in \mathbf{Z}_{\geq 1}$ and $b_k \in \mathbf{N}_{<p}$ with $\sum b_k \leq c$ are arbitrary.

We rephrase this result by introducing the notion of quasi-twist-recurrent functions:

def:16557

Definition 6.2. — We say a function $x: \mathbf{Q} \rightarrow \overline{\mathbf{F}}_p$ is **quasi-twist-recurrent** (QTR) with respect to the data $(a, b, c, M, N) \in \mathbf{Z}_{>0} \times \mathbf{N} \times \mathbf{N} \times \mathbf{Z}_{>0} \times \mathbf{Z}_{>0}$, if its support $\text{Supp}(x)$ is a well-ordered subset of $\mathbf{Q}$ such that

1. for any $q \in \text{Supp}(x)$, if we expand aq in base p as

$$aq = w - 0.q_1 \cdots q_n \cdots, \quad w \in \mathbf{Z}, \quad q_n \in \{0, \dots, p-1\}$$

with $\sum q_n < \infty$, then $w \geq -b$ and $\sum q_n \leq c$;

2. for any integer $w \geq -b$ and any rational number q of the form

$$q = \frac{1}{a} (w - 0.q_1 \cdots q_n \cdots), \quad q_n \in \mathbf{N}_{<p}, \quad \sum q_n \leq c,$$

if there exist M consecutive 0 in the sequence $q_1, q_2, \dots$, i.e. $q_{k+1} = q_{k+2} = \dots = q_{k+M} = 0$ for some integer $k \geq 0$, then the coefficient of x at q equals the coefficient of x at the following rational number:

$$\frac{1}{a} \left(w - 0.q_1 \cdots q_k \overbrace{0 \cdots 0}^{M+N} q_{k+M+1} \cdots \right).$$

We omit the data (a, b, c, M, N) when it is clear from the context, and we say a subset of $\mathbf{Q}$ is QTR if it is the support of a QTR function.

Theorem 6.1 can be restated as follows:

⁽⁴⁾See also [Ked17, Remark 2.7, Remark 2.9] for the critical remarks on [Ked01b, Theorem 15].

prop:54845

Proposition 6.3. — A Hahn series $x = \sum_q x_q t^q \in \overline{\mathbf{F}}_p((t^{\mathbf{Q}}))$ is algebraic over $\overline{\mathbf{F}}_p((t))$ if and only if the function F_x is QTR.

On the other hand, Kedlaya proved the following result, which links the algebraicity of p -adic Hahn series to that of equal-characteristic Hahn series:

thm:47

Theorem 6.4 (cf. [Ked17, Theorem 13.4]). — Let L be the completed integral closure of $\overline{\mathbf{F}}_p((t))$ in $\overline{\mathbf{F}}_p((t^{\mathbf{Q}}))$. Then the completed integral closure of $\check{\mathbf{Q}}_p$ in $\mathbf{L}_p$ is the completion of the following set:

$$\left\{ \sum_q [x_q] t^q \in \check{\mathbf{Z}}_p((t^{\mathbf{Q}})) \mid \sum_q x_q t^q \in L \right\}.$$

Remark 6.5. — Although not directly used in the proof of Theorem 6.23, we note that the completed integral (algebraic) closure of $\check{\mathbf{Q}}_p$ in $\mathbf{L}_p$ is exactly $\mathbf{C}_p$. This is a direct consequence of the fact that $\mathbf{C}_p$ is complete and algebraically closed.

Although the $\mathbf{Q}_p$ -algebraicity (resp. $\check{\mathbf{Q}}_p$ -algebraicity) refinement of Theorem 6.4 is still open, we observe that a finer necessary condition for the $\check{\mathbf{Q}}_p$ -algebraicity of p -adic Hahn series can be derived from Proposition 6.3 and Theorem 6.4:

prop:167

Proposition 6.6. — Let $f = \sum_{q \in \mathbf{Q}} [f(q)] p^q \in \mathbf{L}_p$ be a p -adic Hahn series with bounded support. If f is algebraic over $\check{\mathbf{Q}}_p$, then the function F_f is QTR.

This is a direct corollary of the following lemma:

lem:36014

Lemma 6.7. — For any QTR function $\phi: \mathbf{Q} \rightarrow \overline{\mathbf{F}}_p$, and any integer r , the restriction of ϕ to $(-\infty, r]$ is still QTR.

Proof. — Write ϕ_r for the restriction of ϕ to $(-\infty, r]$, so that $\phi_r(q) = \phi(q)$ for $q \leq r$ and $\phi_r(q) = 0$ for $q > r$; in particular ϕ_r and ϕ agree on $(-\infty, r]$ and $\text{Supp}(\phi_r) = \text{Supp}(\phi) \cap (-\infty, r]$. We check that ϕ_r is QTR with the same data (a, b, c, M, N) as ϕ .

Since $\text{Supp}(\phi_r) \subseteq \text{Supp}(\phi)$ is well-ordered and the first condition of Definition 6.2 only constrains elements of the support, it holds for ϕ_r with the same a, b, c .

For the second condition, fix an integer $w \geq -b$ and a rational number

$$q = \frac{1}{a}(w - 0.q_1 \cdots q_n \cdots), \quad q_n \in \mathbf{N}_{<p}, \quad \sum q_n \leq c,$$

with $q_{k+1} = \cdots = q_{k+M} = 0$ for some integer $k \geq 0$, and set

$$q' := \frac{1}{a} \left(w - 0.q_1 \cdots q_k \overbrace{0 \cdots 0}^{M+N} q_{k+M+1} \cdots \right).$$

We must show $\phi_r(q) = \phi_r(q')$. Inserting N extra zeros shifts the digits $q_{k+M+1}, q_{k+M+2}, \dots$ to the right, so the decimal part of aq' is at most that of aq ; hence $aq' \geq aq$, i.e. $q' \geq q$.

If $q > r$, then $q' \geq q > r$, so $\phi_r(q) = 0 = \phi_r(q')$.

If $q \leq r$, then from $aq = w - 0.q_1 \cdots \leq ar$ together with $0 \leq 0.q_1 \cdots < 1$ and $w, ar \in \mathbf{Z}$ we

get $w \leq ar$; therefore $aq' = w - 0.q_1 \cdots q_k \overbrace{0 \cdots 0}^{M+N} q_{k+M+1} \cdots \leq w \leq ar$, so $q' \leq r$ as well. Then $\phi_r(q) = \phi(q)$ and $\phi_r(q') = \phi(q')$, which are equal because ϕ satisfies the second condition of Definition 6.2.

In either case $\phi_r(q) = \phi_r(q')$, so ϕ_r is QTR. $\square$

Proof of Proposition 6.6. — Suppose that $\text{Supp}(f)$ is bounded from above by some $u \in \mathbf{Z}$. Then Theorem 6.4 shows that there exists $f' \in \mathbf{L}_p$ such that F_f and $F_{f'}$ coincide on $(-\infty, u]$ and f' lies in the completed integral closure of $\overline{\mathbf{F}}_p((t))$. Similarly, there exists $f'' \in \mathbf{L}_p^b$ such that $F_{f'}$

and $F_{f''}$ coincide on $(-\infty, u]$ and $f'' \in \overline{\mathbf{F}}_p((t))^{\text{alg}}$. By Proposition 6.3, the function $F_{f''}$ is QTR, and the result follows from Lemma 6.7. $\square$

sec:23337

6.2. Ray decomposition of the QTR sets. — Although the definitions of QTR sets look complicated, we show in this section that if one imposes restrictions on their order type, then a bounded QTR set can be decomposed into finitely many pieces with a very simple structure, which we call the ray decomposition.

Throughout Section 6.2, we fix a set $S \subseteq S_{a,b,c}$ with the following properties:

it:65165

(S1) $S \subseteq S_{a,b,c,m}$ for some integer $m \geq -b$, where

$$S_{a,b,c,m} = \left\{ \frac{1}{a} \left(m - \sum_{k=1}^{\infty} q_k p^{-k} \right) \mid q_k \in \mathbf{N}_{<p}, \sum q_k \leq c \right\} \subset S_{a,b,c}.$$

it:49436

(S2) There exists $M, N \in \mathbf{N}$ such that if an element $q \in S$ of the form

$$q = \frac{1}{a}(m - 0.q_1 \cdots q_n \cdots), \quad q_n \in \mathbf{N}_{<p}, \sum q_n \leq c$$

satisfies that there exist M consecutive 0 in the sequence $q_1, q_2, \dots$, i.e. $q_{k+1} = q_{k+2} = \dots = q_{k+M} = 0$ for some integer $k \geq 0$, then the following rational number also belongs to S :

$$\frac{1}{a} \left(m - 0.q_1 \cdots q_k \overbrace{0 \cdots 0}^{M+N} q_{k+M+1} \cdots \right).$$

it:31201

(S3) The order type of S is strictly less than ω^2 . In other words, S admits only finitely many accumulation points.

We call such a set S (a, b, c, m, M, N) -admissible.

Remark 6.8. — Note that the condition (S2) is strictly weaker than the one we use to define QTR sets. On the other hand, as shown in the course of the proof of Lemma 6.22, bounded QTR sets are always a finite disjoint union of QTR sets that are contained in some $S_{a,b,c,m}$ for some integer $m \geq -b$. Thus the assumption $S \subseteq S_{a,b,c,m}$ entails no loss of generality when we study the structure of bounded QTR sets.

We begin by defining the concepts of words and gap vectors:

Definition 6.9. —

1. By a **word**, we mean an ordered tuple of elements in $\{1, \dots, p-1\}$ of finite length. For an element $q \in S$ of the form

$$q = \frac{1}{a}(m - 0.q_1 \cdots q_n \cdots),$$

the **word** of q is the tuple of nonzero digits of $0.q_1 \cdots q_n \cdots$.

2. By a **gap vector**, we mean an ordered tuple of natural numbers of finite length. For an element

$$q = \frac{1}{a}(m - 0.q_1 \cdots q_n \cdots)$$

of S , the i -th coordinate of the **gap vector** of q is the number of consecutive 0 between the $(i-1)$ -th and i -th nonzero digits of $0.q_1 \cdots q_n \cdots$. In particular, the first coordinate of the gap vector of q is the number of consecutive 0 before the first nonzero digit of $0.q_1 \cdots q_n \cdots$.

Example 6.10. — Let $q = \frac{1}{a}(m - 0.001104514)$, then the word of q is $(1, 1, 4, 5, 1, 4)$ and the gap vector of q is $(2, 0, 1, 0, 0, 0)$.

Since the value of $0.q_1 \cdots q_n \cdots$ is determined by the positions and the values of its nonzero digits, an element of S is uniquely determined by the pair consisting of its word and its gap vector. We shall freely identify an element of S with this pair; note that the zeros following the last nonzero digit of $0.q_1 \cdots q_n \cdots$ are not recorded by the gap vector.

The following lemma is a direct consequence of (S1):

Lem:5966

Lemma 6.11. — *For any element of S , the length of its word is at most c . In particular, there are only finitely many possible words for elements of S .*

The condition (S2) can be rephrased as follows:

Lem:30089

Lemma 6.12. — *Fix a word $\mathbf{d} = (d_1, \dots, d_t)$ and let $G_{\mathbf{d}} \subseteq \mathbf{N}^t$ be the set of gap vectors of elements of S with word $\mathbf{d}$. If $g \in G_{\mathbf{d}}$ and $g_i \geq M$ for some $i \leq t$, then $g + Ne_i \in G_{\mathbf{d}}$, where e_i is the i -th standard basis vector of $\mathbf{N}^t$.*

We observe that the ordinal bound of S (i.e. condition (S3)) imposes a strong restriction on the gap vectors of elements of S :

Lem:27279

Lemma 6.13. — *For every word $\mathbf{d}$ occurring in S and any gap vector $g \in G_{\mathbf{d}}$, at most one coordinate of g is at least M .*

Proof. — Suppose there exist $i < j$ such that $g_i \geq M$ and $g_j \geq M$. Then by Lemma 6.12, one has $g + sNe_i + tNe_j \in G_{\mathbf{d}}$ for any $s, t \in \mathbf{N}$. We set $v_{s,t}$ to be the element of S with word $\mathbf{d}$ and gap vector $g + sNe_i + tNe_j$. For any fixed s , the sequence $\{v_{s,t}\}_{t \in \mathbf{N}}$ is strictly increasing and consequently has order type ω . On the other hand, for any $s < s'$, one has $v_{s,t} < v_{s',t'}$ for any $t, t' \in \mathbf{N}$. Therefore, the sequence $\{v_{s,t}\}_{s,t \in \mathbf{N}}$ has order type ω^2 , which contradicts the assumption on S . $\square$

Lem:19048

Lemma 6.14. — *For any fixed word $\mathbf{d}$, one can write*

$$G_{\mathbf{d}} = W \cup A_1 \cup \cdots \cup A_r,$$

where W is a finite set, and each A_i has the form $\{v + kNe_u | k = 0, 1, 2, \dots\}$, where e_u is the u -th standard basis vector of $\mathbf{N}^t$ and $v \in G_{\mathbf{d}}$ satisfies $v_j \geq M$ if and only if $j = u$.

Proof. — Take W to be the set of gap vectors in $G_{\mathbf{d}}$ with all coordinates less than M . Since the length of the gap vectors is fixed, and each coordinate of the gap vectors in W has only finitely many possible values, W is a finite set.

By Lemma 6.12, we know that each element of $G_{\mathbf{d}} \setminus W$ has exactly one coordinate that is at least M . Thus

1. There are only finitely many i such that there exists an element of $G_{\mathbf{d}} \setminus W$ whose gap vector has its i -th coordinate at least M .
2. For each i as in (1), there are only finitely many possible values of the k -th coordinate of the gap vector in $G_{\mathbf{d}} \setminus W$ for $k \neq i$.
3. When i and values of other coordinates of the gap vector in $G_{\mathbf{d}} \setminus W$ are fixed, there are only N possibilities for the value of the i -th coordinate of the gap vector in $G_{\mathbf{d}} \setminus W$ modulo N .

Once i , values of other coordinates of the gap vector in $G_{\mathbf{d}} \setminus W$ and the value of the i -th coordinate of the gap vector in $G_{\mathbf{d}} \setminus W$ modulo N are fixed, we denote by $T \subseteq \mathbf{N}^t$ the set of elements of $G_{\mathbf{d}} \setminus W$ with the above fixed data. If T is empty, then it contributes nothing to $G_{\mathbf{d}}$. If T is nonempty, we take $v \in T$ with the smallest value of the i -th coordinate of the gap vector. Then $T \subseteq \{v + kNe_i | k = 0, 1, 2, \dots\}$. The reverse inclusion also holds by Lemma 6.12. $\square$

def:33761

Definition 6.15. — *A ray in S is a subset of S of the form*

$$\left\{ \frac{1}{a} \left(m - 0.q_1 q_2 \cdots q_s \overbrace{0 \cdots 0}^{kN} q_{s+1} \cdots \right) \mid k = 0, 1, \dots \right\},$$

where $\frac{1}{a}(m - 0.q_1q_2 \cdots q_n \cdots)$ is a fixed element of S such that there is one and only one coordinate of the gap vector of it that is at least M .

ray-progression

Remark 6.16. — Under the identification of an element of S with its (word, gap vector) pair, inserting kN zeros at the unique gap of length at least M corresponds to adding kNe_u to the gap vector. Hence, for a fixed word $\mathbf{d}$, the rays in S with word $\mathbf{d}$ are precisely the arithmetic progressions $\{v + kNe_u \mid k \geq 0\}$ furnished by Lemma 6.14. Geometrically, a ray is a strictly increasing sequence converging to its limit point $\frac{1}{a}(m - 0.q_1 \cdots q_s) \in \mathbf{Q}$; this limit point is an accumulation point of S , which explains why the finiteness of the accumulation points (condition (S3)) controls the number of rays.

coro:8081

Corollary 6.17. — The set S can be written as a union of a finite set and finitely many rays.

Proof. — Different decimals $0.q_1q_2 \cdots q_n \cdots$ in Definition 6.15 give rise to disjoint rays. The result follows from Lemma 6.11 and Lemma 6.14. $\square$

In fact, the decomposition in Corollary 6.17 can be further refined to a disjoint union of a finite set and finitely many rays. This is guaranteed by the following lemma:

lem:59667

Lemma 6.18. — Let R_1, R_2 be two rays in S . Then exactly one of the following cases can happen:

1. $R_1 \subseteq R_2$ or $R_2 \subseteq R_1$;
2. $R_1 \cap R_2$ is a finite set.

Proof. — Write

$$R_1 = \left\{ \frac{1}{a}(m - \alpha_1 - \lambda_1 p^{-Nk}) \mid k = 0, 1, \dots \right\}, \quad R_2 = \left\{ \frac{1}{a}(m - \alpha_2 - \lambda_2 p^{-Nk}) \mid k = 0, 1, \dots \right\}.$$

If $\alpha_1 = \alpha_2$, then the condition $q \in R_1 \cap R_2$ yields

$$m - \alpha_1 - aq = \lambda_1 p^{-Nk_1} = \lambda_2 p^{-Nk_2}$$

for some $k_1, k_2 \in \mathbf{N}$. Without loss of generality, we assume $k_1 \leq k_2$. Then one has $\lambda_1 = \lambda_2 p^{-N(k_2 - k_1)}$ and consequently $R_1 \subseteq R_2$.

If $\alpha_1 \neq \alpha_2$ and $R_1 \cap R_2$ is an infinite set, then $R_1 \cap R_2$, as an infinite subsequence of R_1 (resp. R_2), converges to its accumulation point $\frac{1}{a}(m - \alpha_1)$ (resp. $\frac{1}{a}(m - \alpha_2)$), which forces $\alpha_1 = \alpha_2$, a contradiction. $\square$

prop:29055

Proposition 6.19. — The set S can be written as a union of a finite set and finitely many rays that are pairwise disjoint.

Proof. — By Corollary 6.17, we may write $S = E_0 \cup R_1 \cup \cdots \cup R_s$, where E_0 is a finite set and $R_1, \dots, R_s$ are rays. By Lemma 6.18, after discarding every ray that is contained in another one, we may assume that no R_i is contained in another; then $R_i \cap R_j$ is a finite set for all $i \neq j$. Each ray is a strictly increasing sequence parametrized by $k = 0, 1, \dots$ as in Definition 6.15; for $k_0 \geq 0$, let R'_i denote the tail of R_i consisting of the elements with $k \geq k_0$, which is again a ray. Since there are finitely many pairs (i, j) and each $R_i \cap R_j$ is finite, we may choose k_0 large enough that $R'_1, \dots, R'_s$ are pairwise disjoint. Then $\bigcup_{i=1}^s (R_i \setminus R'_i)$ is a finite set, and setting $E := E_0 \cup \bigcup_{i=1}^s (R_i \setminus R'_i)$ yields the pairwise disjoint decomposition $S = E \sqcup R'_1 \sqcup \cdots \sqcup R'_s$. $\square$

coro:48108

Corollary 6.20. — Let U be a bounded QTR set with respect to the data (a, b, c, M, N) . If U has only finitely many accumulation points, then U can be written as a union of a finite set and finitely many pairwise disjoint rays, each contained in some (a, b, c, m, M, N) -admissible subset of U with $m \geq -b$.

Proof. — Since U is bounded, there exist finitely many integers $m_i \geq -b$ such that $U \subseteq \bigcup_i S_{a,b,c,m_i}$. The result follows by applying Proposition 6.19 to each $S_{a,b,c,m_i} \cap U$, which is

(a, b, c, m_i, M, N) -admissible and has finitely many accumulation points, together with the fact that the sets $S_{a,b,c,m_i} \cap U$ are disjoint for different m_i . $\square$

sec:24235

6.3. Sparse representatives and finiteness of bounded QTR supports. —

Lem:57121

Lemma 6.21. — *Let $N \geq 1$ be an integer, and let $\delta_1, \dots, \delta_r$ with $r \geq 1$ be nonzero rational numbers in $[0, 1)$ with finite length base- p expansions. Then the set*

$$W = \{\delta_i p^{-Nk} \mid 1 \leq i \leq r, k = 0, 1, 2, \dots\}$$

is sparse.

Proof. — Let $\delta_i = 0.\delta_{i,1}\delta_{i,2}\dots\delta_{i,s}$ for each i , where s is a large enough integer that does not depend on i . Let

$$C := \max \left\{ \sum_{j=1}^s \delta_{i,j} \mid i = 1, \dots, r \right\}$$

be the maximal digit sum of the δ_i . Since the δ_i are nonzero, one has $C \geq 1$. Without loss of generality, we assume the digit sum of δ_1 is C .

Fix an integer $n \geq 1$. Set $d_i = \delta_1 \cdot p^{-(2s+1)Ni} \in W$ for $i = 1, \dots, n$. Then $d_i \in \text{Dom}_p(W)$ and there is no carry in base p when adding $d_1, \dots, d_n$ together.

Suppose $e_1, \dots, e_n$ are elements of W such that $d_1 + \dots + d_n - (e_1 + \dots + e_n)$ is an integer. Let $\underline{d}^{(i)}$ (resp. $\underline{e}^{(i)}$) be the element in $\mathbb{P}$ such that $\|\underline{d}^{(i)}\| = d_i$ and $\|\underline{e}^{(i)}\| = e_i$. Then

eq:14517

$$(f) \quad \mathfrak{N}_p \left(\sum_i e_i \right) = \Psi \left(\tau \left(\sum_i \underline{e}^{(i)} \right) \right) \leq \Psi \left(\sum_i \underline{e}^{(i)} \right) = \sum_i \Psi \left(\underline{e}^{(i)} \right) \leq n \cdot C = \mathfrak{N}_p \left(\sum_i d_i \right).$$

Since $d_1 + \dots + d_n - (e_1 + \dots + e_n)$ is an integer, we know that $\mathfrak{N}_p(\sum_i e_i) = \mathfrak{N}_p(\sum_i d_i)$ and consequently all inequalities in (f) are equalities. In particular, one has $\mathfrak{N}_p(e_i) = \Psi(\underline{e}^{(i)}) = C$ for each i , and there is no carry in base p when adding $e_1, \dots, e_n$ together. As a result, we know that $\sum_i d_i = \sum_i e_i$.

Note that for any positive integers $i_1 < i_2$, our construction of the d_i ensures that

$$\text{dist} \left(\{j \in \mathbf{Z}_{\geq 1} \mid \underline{d}_j^{(i_1)} \neq 0\}, \{j \in \mathbf{Z}_{\geq 1} \mid \underline{d}_j^{(i_2)} \neq 0\} \right) > s.$$

Fix $l \in \mathbf{Z}_{\geq 1}$. Since $\text{diam}(\{j \in \mathbf{Z}_{\geq 1} \mid \underline{e}_j^{(l)} \neq 0\})$ does not exceed s , we conclude that the set

$$\left\{ i \mid \exists w \in \mathbf{Z}_{\geq 1}, \underline{d}_w^{(i)} \neq 0 \wedge \underline{e}_w^{(l)} \neq 0 \right\}$$

has at most one element, i.e. the digits of e_l overlap with those of d_i for at most one i .

On the other hand, the conditions

1. there is no carry in base p when adding $d_1, \dots, d_n$ (resp. $e_1, \dots, e_n$) together;
2. $d_1, \dots, d_n$ (resp. $e_1, \dots, e_n$) have the same digit sum C ;
3. $\sum_i d_i = \sum_i e_i$;

ensures that e_l is identical to certain d_i . We conclude that $e_1, \dots, e_n$ are just a permutation of $d_1, \dots, d_n$. This shows that W is sparse. $\square$

Lem:42556

Lemma 6.22. — *Let $f \in \mathbf{L}_p$ be a $\check{\mathbf{Q}}_p$ -algebraic p -adic Hahn series such that the support is bounded and admits finitely many (at least one) accumulation points. Then there exists $S' \subseteq \text{Supp}(f)$, $\lambda \in \mathbf{Q}$ and an integer $T \geq 1$ such that $\text{Supp}(f) \setminus S'$ is a finite set and $-T(S' - \lambda)$ admits a sparse nonzero set of representatives modulo $\mathbf{Z}$.*

Proof. — By Proposition 6.6, we know that F_f is QTR for certain a, b, c, M, N . This implies that $\text{Supp}(f) \subseteq S_{a,b,c}$. Since $\text{Supp}(f)$ is bounded, there exists a finite index set $I \subseteq \mathbf{Z} \cap [-b, \infty)$ such that $\text{Supp}(f) \subseteq \bigcup_{m \in I} S_{a,b,c,m}$. For every $m \in I$, we set $S_m = \text{Supp}(f) \cap S_{a,b,c,m}$. Then S_m is (a, b, c, m, M, N) -admissible. By Proposition 6.19, S_m can be written as the pairwise disjoint union of a finite set and finitely many rays in S_m . Since the rays are also disjoint across different m , we set $S'' = \bigsqcup_{l=1}^s R_l$ to be the (nonempty) union of all rays in S_m for all $m \in I$. Then $\text{Supp}(f) \setminus S''$ is a finite set, and S'' is a disjoint union of finitely many rays. For each ray R_l , we write

$$R_l = \{a^{-1}(m_l - \alpha_l - \beta_l p^{-Nk}) \mid k \in \mathbf{N}\},$$

where $m_l \geq -b$ is an integer, $\alpha_l \in [0, 1) \cap \mathbf{Q}$, $\beta_l \in (0, 1) \cap \mathbf{Q}$, and α_l, β_l have finite base- p expansions. Each ray has rational limit point $\lambda_l := a^{-1}(m_l - \alpha_l)$.

Choose an integer $T \geq 1$ such that a divides T and $T(\lambda_l - \lambda_1) \in \mathbf{Z}$ for every $l = 1, 2, \dots, s$. Then for any $l = 1, \dots, s$ and any $q = a^{-1}(m_l - \alpha_l - \beta_l p^{-Nk}) \in R_l$,

$$-T(q - \lambda_1) = -T(\lambda_l - \lambda_1) + \frac{T}{a}\beta_l p^{-Nk} \equiv \gamma_l p^{-Nk} \pmod{\mathbf{Z}},$$

where $\gamma_l = \frac{T}{a}\beta_l$ is a nonzero rational number. Since $\frac{T}{a}$ is an integer and β_l has finite base- p expansion, γ_l has finite base- p expansion, and hence $\gamma_l \in \mathbf{Z}[1/p]$. Choose a large enough integer $K \geq 0$ such that $0 < \gamma_l p^{-NK} < 1$ for every $l = 1, \dots, s$. We set $\delta_l = \gamma_l p^{-NK}$ for every $l = 1, \dots, s$.

We delete the first K points of each ray in S'' and denote the resulting set by S' , i.e.

$$S' = \left\{a^{-1}(m_l - \alpha_l - \beta_l p^{-N(j+K)}) \mid l = 1, \dots, s, j \in \mathbf{N}\right\}.$$

Since this is a finite deletion, $\text{Supp}(f) \setminus S'$ is still a finite set. On the other hand, for any $q = a^{-1}(m_l - \alpha_l - \beta_l p^{-N(j+K)}) \in S'$, one has

$$-T(q - \lambda_1) \equiv \delta_l p^{-Nj} \pmod{\mathbf{Z}}.$$

Conversely, for any $l = 1, \dots, s$ and any $j \in \mathbf{N}$, there exists $q = a^{-1}(m_l - \alpha_l - \beta_l p^{-N(j+K)}) \in S'$ such that

$$-T(q - \lambda_1) \equiv \delta_l p^{-Nj} \pmod{\mathbf{Z}}.$$

Thus the elements $\delta_l p^{-Nj}$ for $l = 1, \dots, s$ and $j = 1, 2, \dots$, after deleting the duplicates, form a nonzero set of representatives of $-T(S' - \lambda_1)$ modulo $\mathbf{Z}$, which is sparse by Lemma 6.21. $\square$

thm:29075

Theorem 6.23. — *Let f be a p -adic Hahn series satisfying the following conditions:*

1. *f is algebraic over $\check{\mathbf{Q}}_p$;*
2. *$\text{Supp}(f)$ is bounded and admits finitely many accumulation points.*

Then $\text{Supp}(f)$ is a finite set.

Proof. — Suppose that $\text{Supp}(f)$ is an infinite set. Then there exists at least one accumulation point of $\text{Supp}(f)$.

By Lemma 6.22, we can write $\text{Supp}(f) = W \sqcup S'$, where W is a finite set, and S' is an infinite set such that $-T(S' - \lambda)$ admits a nonzero sparse set of representatives modulo $\mathbf{Z}$ for some $\lambda \in \mathbf{Q}$ and integer $T \geq 1$. We write $f = f_0 + f_1$ according to the decomposition $\text{Supp}(f) = W \sqcup S'$, i.e., f_0 (resp. f_1) is the p -adic Hahn series consisting of the terms of f with support in W (resp. S'). Since W is a finite set, f_0 is algebraic over $\check{\mathbf{Q}}_p$. This forces f_1 , and consequently $f_1 \cdot p^{-\lambda}$, to be algebraic over $\check{\mathbf{Q}}_p$. Note that the support of $f_1 \cdot p^{-\lambda}$ is just $S' - \lambda$, and Theorem 5.3 implies that $f_1 \cdot p^{-\lambda}$ is transcendental over $\check{\mathbf{Q}}_p$, a contradiction. $\square$

coro:11594

Corollary 6.24. — *Let $f \in \mathbf{L}_p$ be a p -adic algebraic number. Then $\text{Supp}(f)$ has either no accumulation points or infinitely many accumulation points. In other words, its order type is either finite or not less than ω^2 .*

sec:formal

Appendix A. AI-assisted formalization in Lean 4

A.1. Overview of the formalization project. — The formalization of this paper, which we refer to as the `FormalizedSparse` project, contains approximately 24,000 lines of Lean code (including docstrings) and is organized into the following files:

A.1.1. `WittVector.lean`. — This file contains our realization of $\check{\mathbf{Q}}_p$, the completed maximal unramified extension of $\mathbf{Q}_p$, and its ring of integers $\check{\mathbf{Z}}_p$.

Although Mathlib already contains a formalization of $\mathbf{Q}_p$, the lack of infrastructure on ramification theory in Mathlib makes it difficult to define $\check{\mathbf{Q}}_p$ literally as the union of all finite unramified extensions of $\mathbf{Q}_p$. Instead, we start with $\check{\mathbf{Z}}_p$, which we define as the ring of Witt vectors over $\overline{\mathbf{F}}_p$:

```
-- The algebraic closure of ' $\mathbf{F}_p$ '
abbrev Fpbar (p : ℕ) [Fact (Nat.Prime p)] := AlgebraicClosure (ZMod p)
notation "Fa_" p " " => Fpbar p

-- The ring of integers of completed maximal unramified extension of ' $\mathbf{Q}_p$ ',
abbrev OQpUn (p : ℕ) [Fact (Nat.Prime p)] := WittVector p (Fpbar p)
notation "Zu_" p " " => OQpUn p
```

Then we define $\check{\mathbf{Q}}_p$ as the fraction field of $\check{\mathbf{Z}}_p$, with the induced valuation:

```
abbrev QpUn (p : ℕ) [Fact (Nat.Prime p)] :=
  WithVal ((IsDiscreteValuationRing.maximalIdeal (Zu_[p])).valuation
    ((FractionRing (Zu_[p]))))
notation "Qu_" p " " => QpUn p
```

We establish several `instance` around $\mathbb{Q}^u_{-}[p]$, such as the fact that it is a complete rank-1 valued field. Besides that, we define the embedding from $\mathbf{Q}_p$ to $\check{\mathbf{Q}}_p$: one has $\mathbf{Z}_p \cong W(\mathbf{F}_p)$, which injects into $\check{\mathbf{Z}}_p := W(\overline{\mathbf{F}}_p)$. This extends to an embedding $\mathbf{Q}_p \hookrightarrow \check{\mathbf{Q}}_p$:

```
-- The embedding from  $\mathbb{Q}_{-}[p]$  to  $\mathbb{Q}^u_{-}[p]$ .
noncomputable def Qp_embd {p : ℕ} [Fact (Nat.Prime p)] :  $\mathbb{Q}_{-}[p] \rightarrow^{**} \mathbb{Q}^u_{-}[p]$  :=
  @IsFractionRing.map Z-[p] Zu_-[p]  $\mathbb{Q}_{-}[p]$   $\mathbb{Q}^u_{-}[p]$  - - - - -
  ((WittVector.map (algebraMap (ZMod p) (Fa_-[p]))).comp
    (WittVector.fromPadicInt p)) ... -- The map  $\mathbf{Z}_p \rightarrow \check{\mathbf{Z}}_p$  is injective. Omitted.
```

Finally, we formalize a lemma to show that this embedding preserves the valuation.

A.1.2. `PAdicHahnSeries.lean`. — This file contains the formalization of $\mathbf{L}_p$, the field of p -adic Hahn series. The material is mostly taken from [Poo93].

Since the equal-characteristic Hahn series is already available in Mathlib, we formalize the ring $\check{\mathbf{Z}}_p((t^{\mathbf{Q}}))$ as:

```
abbrev LiftedPAdicHahnSeries (p : ℕ) [Fact (Nat.Prime p)] := HahnSeries  $\mathbb{Q}$  (Zu_-[p])
```

We define the null series condition (cf. Proposition 2.3) as a predicate `IsNullSeries` on `LiftedPAdicHahnSeries`, and define the ideal $\mathcal{N}$ as the set of null series:


```
def NullSeriesIdeal (p : ℕ) [Fact (Nat.Prime p)] : Ideal (LiftedPAdicHahnSeries p)
  ↪ where
    carrier := {x | IsNullSeries x}
    ... -- Omitted.
```

We provide an `instance` to show that $\mathcal{N}$ is a maximal ideal of $\check{\mathbf{Z}}_p((t^{\mathbb{Q}}))$, so that the quotient $\check{\mathbf{Z}}_p((t^{\mathbb{Q}}))/\mathcal{N}$ is a field (cf. (1) of Proposition 2.3).

Instead of directly defining $\mathbf{L}_p$ as the quotient of $\check{\mathbf{Z}}_p((t^{\mathbb{Q}}))$ by $\mathcal{N}$, we first formalize (2) of Proposition 2.3:

```
-- Every element of  $\check{\mathbf{Z}}_p((t^{\mathbb{Q}}))/\mathcal{N}$  has a unique lift of the form  $\sum_{q \in \mathbb{Q}} [f(q)]t^q$ ,
-- with the support a well-ordered set.
theorem exists_canonical_expansion {p : ℕ} [Fact (Nat.Prime p)] :
  ∀ A : (LiftedPAdicHahnSeries p) / (NullSeriesIdeal p),
    ∃! (s : {f : ℚ → ℱ^a_[-p] // (Function.support f).IsPWO}),
      Ideal.Quotient.ringCon (NullSeriesIdeal p)
        A.out (LiftedPAdicHahnSeries.from_coeff s.val s.prop)
```

This allows us to define a valuation on $\check{\mathbf{Z}}_p((t^{\mathbb{Q}}))/\mathcal{N}$ by considering the minimum of the support of this unique lift:

```
def val (p : ℕ) [Fact (Nat.Prime p)] :
  AddValuation ((LiftedPAdicHahnSeries p) / (NullSeriesIdeal p)) (WithTop ℚ) where
  toFun x :=
    if h : x = 0 then (τ : WithTop ℚ) -- If  $x = 0$ , then its valuation is  $\infty$ .
    else ((support_IsPWO x).isWF.min (support_nonempty_of_nonzero p x h) : WithTop ℚ)
    ... -- Omitted.
```

And finally we define $\mathbf{L}_p$ as the fraction field of $\check{\mathbf{Z}}_p((t^{\mathbb{Q}}))/\mathcal{N}$:

```
abbrev pAdicHahnSeries (p : ℕ) [Fact (Nat.Prime p)] : Type _ :=
  (LiftedPAdicHahnSeries p) / (NullSeriesIdeal p)
notation "L_[" p "]" => pAdicHahnSeries p
```

Several facilities around $\mathbf{L}_p$, such as its support and the coefficients (as a function of type $\mathbb{Q} \rightarrow \mathbb{F}^a_[-p]$), are also formalized in this file.

To deliver the $\mathbf{Q}_p$ -transcendence and $\check{\mathbf{Q}}_p$ -transcendence results, we provide the embedding from $\check{\mathbf{Q}}_p$ to $\mathbf{L}_p$. It is induced by the embedding from $\check{\mathbf{Z}}_p$ to $\mathbf{L}_p$, which maps an element w to the image of $w \cdot t^0$ in the quotient $\check{\mathbf{Z}}_p((t^{\mathbb{Q}}))/\mathcal{N}$.

```
def ZpUn_embd {p : ℕ} [Fact (Nat.Prime p)] : ℤ^u_[-p] →+* L_[-p] where
  toFun a := Ideal.Quotient.mk (NullSeriesIdeal p) (HahnSeries.single 0 a)
  ... -- Omitted.

def QpUn_embd {p : ℕ} [Fact (Nat.Prime p)] : ℚ^u_[-p] →+* L_[-p] :=
  IsFractionRing.map (j := ZpUn_embd (p := p)) ZpUn_embd_injective
```

Finally, we show by induction on the cardinality of the support that an element of $\mathbf{L}_p$ with finite support must be algebraic over $\mathbf{Q}_p$:

```
lemma alg_of_fin_supp (p : ℕ) [Fact (Nat.Prime p)] (f : L_[-p])
  (hf : f.support.Finite) : IsAlgebraic ℚ_[-p] f
```


A.1.3. `Tscaled.lean`. — This file corresponds to Section 4 of this paper. We use $\mathbb{Z}^{un}_{p,T}$ (resp. $\mathbb{Q}^{un}_{p,T}$) to denote the ring $\tilde{\mathbb{Z}}_{p,T}$ (resp. $\tilde{\mathbb{Q}}_{p,T}$), and use $\mathbb{L}_{p,T}$ to denote the quotient of $\tilde{\mathbb{Z}}_{p,T}((t^{\mathbb{Q}}))$ by $\mathcal{N}_T$. After we formalize Lemma 4.7 and Lemma 4.8, the isomorphism σ in Proposition 4.9 can be delivered:

```
def σ : L_[p] ≈+ L_[p,T] := ... -- Omitted.

-- Proposition 4.9, σ is given by  $\sum_{q \in \mathbb{Q}} [f(q)]p^q \mapsto \sum_{q \in \mathbb{Q}} [f(q)]p^q$ .
theorem σ_coeff_compat (f : L_[p]) : (σ p T f).coeff = f.coeff
```

A.1.4. `Sparse.lean`. — This file corresponds to Definition 1.3, Definition 1.4 and Section 3 of this paper, which is about the sparseness, (c, n) -sparseness and the related infrastructure.

Fortunately, Mathlib already contains the function `Real.digits`, which extracts the digits of a real number in a given base. This allows us to formalize the p -digit sum of a rational number, and consequently $\text{dom}_p(S)$ and $\text{Dom}_p(S)$ of a set $S \subseteq \mathbb{Q}$ with a little effort:

```
abbrev decDigits (p : ℕ) [Fact (Nat.Prime p)] (q : ℚ) : ℕ+ → Fin p :=
  fun n => Real.digits (Int.fract q) p ((n : ℕ) - 1)

/- The 'p-digit sum', 'Σ_p(q)' in Definition 1.3 (1)-/
def pDigitSum (p : ℕ) [Fact (Nat.Prime p)] (q : ℚ) : WithTop ℕ :=
  if h : (Function.support (decDigits p q)).Infinite then τ
  else ∑ n ∈ (Set.not_infinite.1 h).toFinset, (decDigits p q n).val

/- 'dominant p-digit sum' of S, Definition 1.3 (2)-/
def dom (p : ℕ) [Fact (Nat.Prime p)] (S : Set ℚ) : WithTop ℕ :=
  sSup {pDigitSum p q | q ∈ S}

/- 'p-digit dominant part' of S, Definition 1.3 (2)-/
def Dom (p : ℕ) [Fact (Nat.Prime p)] (S : Set ℚ) : Set ℚ :=
  {q ∈ S | pDigitSum p q = dom p S}
```

With these preparations, we present the formalized version of Definition 1.4:

```
/- The sparse condition, as a predicate -/
def IsSparse (p : ℕ) [Fact (Nat.Prime p)] (S : Set ℚ) : Prop :=
  S ⊆ Set.Ico 0 1 ∧ dom p S < τ ∧
  ∃ D : Set ℕ+, D.Infinite ∧ (∀ n ∈ D, ∃ d : Fin n → Dom p S, (
    ∀ i : ℕ+, ∑ (j : Fin n), (decDigits p (d j) i).val < p
    -- No carrying when adding d_1, d_2, ..., d_n together.
  )) ∧ (∀ e : Fin n → Dom p S,
    (∑ i, (d i).val - ∑ i, (e i).val).isInt →
    ∃ perm : Equiv.Perm (Fin n), ∀ i, d i = e (perm i)
    -- Unique up to a permutation of {1, ..., n}.
  ))
```

Since Section 3 contains mostly implementation-level details, we will not demonstrate most of the formalization here, except for Example 3.7: the p -digit disjoint subset of $[0, 1) \cap \mathbb{Q}$ is sparse:

```
lemma IsSparse_of_digit_disjoint (p : ℕ) [Fact (Nat.Prime p)] (A : ℕ → Set ℕ+)
(hA1 : ∀ n, (A n).Nonempty) (hA2 : ∀ i j, (A i) ∩ (A j) ≠ ∅ → i = j)
(hA3 : ∀ n, (A n).Finite)
(hAsup : ∃ K : ℕ, (∀ n, (hA3 n).toFinset.card ≤ K) ∧
  {n | (hA3 n).toFinset.card = K}.Infinite) :
  IsSparse p {∑ r ∈ (hA3 i).toFinset, (p : ℚ) ^ (-(r : ℤ)) | i : ℕ }
```

A.1.5. MainTheorem.lean. — The single objective of this file is to formalize the proof of Theorem 1.7:

```
theorem main_theorem (p : ℕ) [Fact (Nat.Prime p)] (f : ℒ[p]) (T : ℕ+)
(W : Set ℚ) (hw1 : W ≠ {0}) (hw2 : IsSparse p W)
(hf : IsRepModZ W {-1 * T * q | q ∈ f.support}) :
  ¬ IsAlgebraic ℚun[p] f
```

Here `IsRepModZ` is a predicate to express the condition that W is a set of representatives of $-T \cdot \text{Supp}(f)$ modulo $\mathbb{Z}$.

A variant of this theorem, which replaces the algebraicity over $\check{\mathbb{Q}}_p$ by the algebraicity over $\mathbb{Q}_p$, is also formalized in this file.

A.1.6. QuasiTwistRecurrent.lean. — This file corresponds to Section 6.1. The key concept is that of a quasi-twist-recurrent (QTR) function (Definition 6.2). Throughout, the base- p digit string $0.q_1 \cdots q_n \cdots = \sum_{i \geq 1} q_i p^{-i}$ is modeled by a finitely supported $d : \mathbb{N} \rightarrow_0 \mathbb{N}$, where $d\ i$ is the digit q_{i+1} . It is implemented as a predicate as follows:

```
def IsQTR {p : ℕ} [Fact (Nat.Prime p)] (x : ℚ → ℱa[p])
(a : ℕ+) (b c : ℕ) (M N : ℕ+) : Prop :=
  (Function.support x).IsWF ∧ -- The support is a well-ordered subset of ℚ.
  (Function.support x ⊆ Kedlaya.Sabc p a b c) ∧ -- The support is contained in Sa,b,c.
  -- The M-zero-run ⇒ N-zero-insertion recurrence.
  ∀ (w : ℤ), -(b : ℤ) ≤ w → ∀ (d : ℕ →0 ℕ), (∀ i, d i < p) →
    (d.sum fun _ v => v) ≤ c →
    ∀ (k : ℕ), (∀ i, k ≤ i → i < k + (M : ℕ) → d i = 0) →
      x ((1 / (a : ℚ)) * ((w : ℚ) - d.sum fun i v =>
        (v : ℚ) * (p : ℚ) ^ (-(i + 1 : ℤ)))) = x ((1 / (a : ℚ)) * ((w : ℚ) -
        (Finsupp.mapDomain (fun i =>
          if i < k + (M : ℕ) then i
          else i + (N : ℕ)) d).sum fun i v => (v : ℚ) * (p : ℚ) ^ (-(i + 1 : ℤ)))))
```

Here `Kedlaya.Sabc p a b c` is the set $S_{a,b,c}$ of Theorem 6.1, and the insertion of N zeros into a length- M zero gap is realized by `Finsupp.mapDomain`. Building on Kedlaya's criterion (Theorem 6.1 and 6.4), the main result of this file is the necessary QTR condition for bounded $\mathbb{Q}_p$ -algebraic p -adic Hahn series (Proposition 6.6):

```
theorem isQTR_of_isAlgebraic_of_bddSupport {p : ℕ} [Fact (Nat.Prime p)]
(f : ℒ[p]) (halg : IsAlgebraic ℚun[p] f) (hbdd : IsBounded f.support) :
  ∃ (a : ℕ+) (b c : ℕ) (M N : ℕ+), IsQTR f.coeff a b c M N
```

A.1.7. RayDecomposition.lean. — This file corresponds to Section 6.2. The central concept is that of a ray (Definition 6.15). Using the same `finsupp` digit model as above, a ray with base element `d_base` is obtained by inserting kN zeros at a fixed gap position `shift_pos` for $k = 0, 1, 2, \dots$. It is formalized as a predicate as follows:

```
def IsRay (p : ℕ) [Fact (Nat.Prime p)] (a : ℕ+) (c : ℕ) (m : ℤ) (N : ℕ+) (S : Set ℚ)
(R : Set ℚ) : Prop := R ⊆ S ∧
  ∃ (d_base : ℕ →0 ℕ) (shift_pos : ℕ),
    (∀ i, d_base i < p) ∧ (d_base.sum fun _ v => v) ≤ c ∧
    R = { q : ℚ | ∃ k : ℕ,
      q = (1 / (a : ℚ)) * ((m : ℚ) -
        (Finsupp.mapDomain
          (fun i => if i < shift_pos then i else i + k * (N : ℕ)) d_base).sum
          fun i v => (v : ℚ) * (p : ℚ) ^ (-(i + 1 : ℤ))) } }
```

The main result of this file is the ray decomposition of bounded QTR sets (Corollary 6.20): a bounded QTR set with finitely many accumulation points is a union of a finite set and finitely many pairwise disjoint rays. Accumulation points are formalized via `derivedSet`, and `rays.sup id` denotes the union of the finite family `rays`:

```
/- Ray decomposition of bounded QTR sets -/
theorem qtr_ray_decomposition {p : ℕ} [Fact (Nat.Prime p)]
{a : ℕ+} {b c : ℕ} {M N : ℕ+} {x : ℚ → ℱa[p]}
(hqtr : IsQTR x a b c M N) (hbdd : Bornology.IsBounded (Function.support x))
(hfin_acc : (derivedSet (Function.support x)).Finite) :
  ∃ (E : Finset ℚ) (rays : Finset (Set ℚ)),
    (∀ R ∈ rays, ∃ (m : ℤ) (hm : -(b : ℤ) ≤ m),
      IsRay p a c m N (Function.support x) R ∧
      IsAdmissible p a (b : ℤ) c m hm M N (Function.support x ∩ Sabc_m p a c m)) ∧
    (∀ R1 ∈ rays, ∀ R2 ∈ rays, R1 ≠ R2 → Disjoint R1 R2) ∧
    Function.support x = ↑E ∪ rays.sup id
```

Here `IsAdmissible` and `Sabc_m` are the admissibility conditions (S1)–(S3) and the m -slice $S_{a,b,c,m}$ from Section 6.2, whose formalization we omit.

A.1.8. BoundedSupport.lean. — This file corresponds to the subsection Section 6.3. Combining the ray decomposition with the sparseness of the explicit witness set (the formalized version of Lemma 6.21) and the main transcendence theorem `main_theorem` (Theorem 5.3), we formalize the finiteness of bounded QTR supports (Theorem 6.23):

```
theorem finite_support_of_qpun_algebraic_of_bounded_support
{p : ℕ} [Fact (Nat.Prime p)]
(f : ℒ[p]) (hf1 : IsAlgebraic ℚun[p] f) (hf2 : IsBounded f.support)
(hf3 : (derivedSet ((Rat.cast : ℚ → ℝ) '' f.support)).Finite) :
  f.support.Finite
```

And consequently Corollary 6.24:

```
-- Qp-algebraic + bounded support ⇒ card{accumulation points} = 0 or ∞.
theorem support_accpt_empty_or_infinite_of_qp_algebraic_of_bounded_support
{p : ℕ} [Fact (Nat.Prime p)]
(f : ℒ[p]) (hf1 : IsAlgebraic ℚ[p] f) (hf2 : IsBounded f.support) :
  (derivedSet ((Rat.cast : ℚ → ℝ) '' f.support)) = ∅ ∨
  (derivedSet ((Rat.cast : ℚ → ℝ) '' f.support)).Infinite

-- The order type variant.
theorem order_type_of_qp_algebraic_of_bounded_support {p : ℕ} [Fact (Nat.Prime p)]
(f : ℒ[p]) (hf1 : IsAlgebraic ℚ[p] f) (hf2 : IsBounded f.support) :
  typeLT f.support < omega0 ∨ typeLT f.support ≥ omega0^2
```

We also formalize a litegral version of Theorem 1.10 in this file, which we omit here for brevity.

A.1.9. HuangStefanescu.lean. — This file contains the formalization of two quick corollary : Corollary 1.12 and Proposition 1.14 of Theorem 1.10.

```
--  $\mathbb{Q}_p$ -algebra + strictly monotone support  $\Rightarrow$  unbounded support
theorem tendsto_atTop_of_strictMono_support_of_qpun_algebraic
{p : ℕ} [Fact (Nat.Prime p)] (s : ℕ → ℚ) (hs : StrictMono s) (f : L_[p])
(hf1 : IsAlgebraic ℚun_[p] f) (hf2 : f.support = Set.range s) :
  Tendsto s atTop atTop

-- The  $p$ -adic analogue of the result of Huang and Stefanescu.
theorem padic_huang_stefanescu_tfae (p : ℕ) [Fact (Nat.Prime p)]
(f : L_[p]) (hf : f.support ⊆ {-(p : ℚ) ^ (-(i : ℤ)) | i : ℕ+}) :
  List.TFAE [f.support.Finite, IsAlgebraic ℚun_[p] f, IsAlgebraic ℚ_-[p] f]
```

A.1.10. *Kedlaya.lean*. — This file records the deep results of Kedlaya (Theorem 6.1 and 6.4) that we use as black boxes. Reproving them is outside the scope of this formalization, so their statements are given here and their proofs are the only intentional **admits** in the project. The shared dependency is the support set $S_{a,b,c}$ of Theorem 6.1, where the base- p digit sequence is again modeled by a $\text{finsupp } d : \mathbb{N} \rightarrow_{\mathfrak{o}} \mathbb{N}$:

```
/- [Ked17, Definition 2.1] -/
def Sabc (a : ℕ+) (b c : ℕ) : Set ℚ :=
{ s : ℚ | ∃ (n : ℤ) (d : ℕ →o ℕ),
  -b ≤ n ∧ (∀ i, d i < p) ∧ (d.sum fun _ v => v) ≤ c ∧
  s = (1 / (a : ℚ)) *
    ((n : ℚ) - d.sum fun i v => (v : ℚ) * (p : ℚ) ^ (-(i + 1 : ℤ))) } }
```

The integrality criterion (Theorem 6.1) is then stated as follows, where $(\mathbb{F}^a_-[p])\langle X \rangle$ is the Laurent series field $\mathbb{F}_p((t))$; note that integrality and algebraicity coincide over a field, so we phrase downstream results in terms of algebraicity:

```
/- [Ked17, Theorem 11.11], [Ked01b, Theorem 15] -/
theorem kedlaya_2001a_theorem15 (x : HahnSeries ℚ (F^a_-[p])) :
  IsIntegral (F^a_-[p])\langle X \rangle x ↔ ∃ a : ℕ+, ∃ b c : ℕ,
  (x.support ⊆ Sabc p a b c) ∧ -- x is supported on some  $S_{a,b,c}$ .
  (∃ M N : ℕ+, ...) -- the twist functions  $f_m$  are eventually periodic, omitted.
:= by admit
```

The second black box is Kedlaya’s description of the completed integral closure of $\check{\mathbb{Q}}_p$ in $\mathbb{L}_p$ (Theorem 6.4), which is the input to Proposition 6.6:

```
/- [Ked17, Theorem 13.4] -/
theorem kedlaya_2017_theorem13_4 :
  closure (integralClosure ℚun_[p] L_[p]).carrier =
  closure { f : L_[p] | ∃ f' : HahnSeries ℚ (F^a_-[p]), IsAlgebraic F^a_-[p] f' ∧
    (exists_canonical_expansion f).choose.val = f'.coeff } := by admit
```

A.2. AI-assisted formalization. — The agentic auto-formalization system Archon, which is based on Claude Opus 4.8 and developed by the AI4Math team of BICMR of Peking University, greatly accelerated our formalization process. By design, once the blueprint or the corresponding informal proof of the project is provided, Archon works fully autonomously to complete the project-level informal-to-formal translation of the statements (as well as mathematical definitions) and to formalize the proof.

A.2.1. *Good translation of the statements and definitions.* — Specific to our project, our experience suggests that it is better to write all the formalized definitions and statements by a

human mathematician with solid experience in Lean⁽⁵⁾, and then let Archon fill in the proofs. While Archon is capable of independently formalizing mathematical concepts, it, like other artificial intelligence systems, lacks an adequate mathematical intuition to formulate definitions in a manner that facilitates their subsequent application.

For example, at the beginning of the project, we let Archon formalize $\check{\mathbf{Q}}_p$, the completed maximal unramified extension of $\mathbf{Q}_p$. As we observed, Archon tried very hard to literally formalize the definition of $\check{\mathbf{Q}}_p$, without success. In fact, the mathematical insight here is that the ramification information is not what we really need. Instead, the key property of $\check{\mathbf{Q}}_p$ that is repeatedly used in this project is that every element of $\check{\mathbf{Q}}_p$ can be uniquely written as a p -adic Laurent series whose coefficients are Teichmüller representatives. This is the perfect scenario for applying the ring of Witt vectors, which is already formalized in Mathlib.

A.2.2. Automatic formalization: proof and revision. — After the formalized statements and definitions are settled⁽⁶⁾, we let Archon fill in the proofs unsupervisedly. The result is quite satisfactory: Archon is able to complete most proofs in a way that closely follows the informal proof, and the generated code (as well as the docstrings) is mostly readable. We learned several things from this process:

1. During its work, the formalization by Archon helped us find several mistakes (which are now fixed) in the original version of the informal proof:
 - (a) Theorem 1.7 does not hold for the sparse set $W = \{0\}$.
 - (b) We did not require the supremum in Example 1.6 to be attained by infinitely many i in the original version, which is also necessary for the proof to work.
 - (c) As we mentioned at the beginning of Section 4, some elements in $\mathbf{L}_p$ were casually written in the form $\sum_{q \in [0, 1/T)} c_q p^q$, with $c_q \in \check{\mathbf{Q}}_{p,T}$ for every q . This is not a well-defined element of $\mathbf{L}_p$, unless we consider the T -scaled realization (i.e., $\mathbf{L}_{-}[\mathbf{p}, \mathbf{T}]$) of $\mathbf{L}_p$ and the isomorphism σ in Proposition 4.9.

These mistakes, which are all fixed in the current version, are all related to the technical details of the proof, and would not have been easy to find without the help of formalization. Archon records these mistakes in the docstrings. For the first two mistakes, Archon added the necessary condition by itself and continued the formalization process without any human intervention. For the third mistake, Archon failed to complete the formalization and terminated with a detailed report.

2. Archon’s ability to backtrack is impressive. When formally proving Theorem 1.7, which is highly combinatorial, we gave no hint to Archon about the structure of the proof, except for the informal proof itself. Archon made multi-level plans to divide the proof into several lemmas and assemble them together to complete the proof. During the work of Archon, we observed that some of its intermediate lemmas were not correct, and Archon was able to backtrack and revise the proof plan to fix the mistake without any human intervention.

A.3. How formalization helps mathematical research. — For mathematicians with limited experience in formalization, agent-based systems such as Archon may eventually provide a practical way to validate proofs in a largely black-box manner. One can envision a future workflow in which a paper written in natural language is automatically translated into a formal proof object and subsequently verified by the system after extensive computation.

⁽⁵⁾In the `FormalizedSparse` project, we formalized all the definitions and statements by hand.

⁽⁶⁾We also built up several helper results (with `sorry`) that we expected to be useful during the formalization process.

For mathematicians with some experience in formalization, we believe that mathematical research can benefit substantially from a human-in-the-loop workflow. In such a workflow, researchers formulate definitions and statements of intermediate results in formal language, while AI systems assist with the labor-intensive formalization process. Successfully formalized intermediate results then provide verified foundations for subsequent arguments, while failed formalizations may help detect errors at an early stage of the research process.

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